IT5352 PROGRAMMING AND DATA STRUCTURES ASSIGNMENT

SUBMITTED BY

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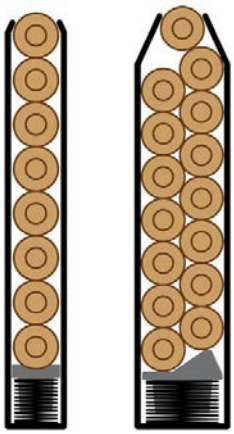
APPLICATIONS OF STACK

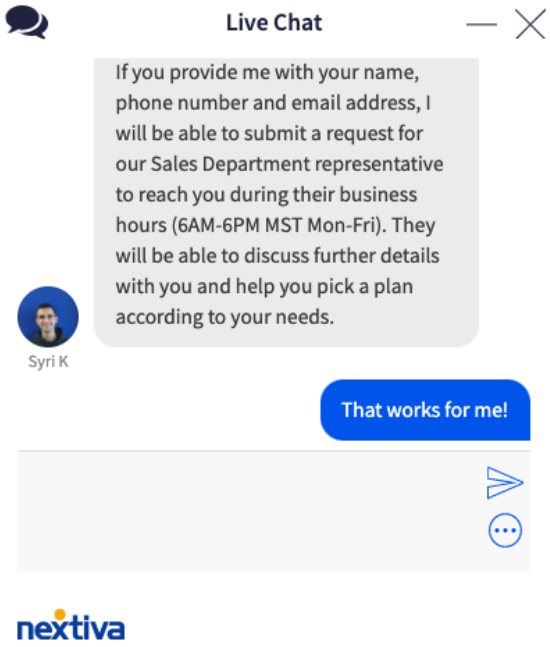
1) Firearms Magazine



A firearms magazine consists of a number of bullets. It works as a stack because bullets can only be inserted from the top, and only the topmost bullet is accessible. The various stack operations are also applicable.

Push: Add a new bullet to the top
Pop: Fire the topmost bullet
Peek: Check if the topmost bullet is correctly placed
isEmpty: Check if there are any bullets left to fire
isFull: Check if the magazine is full

2) Live Chats



Live Chats are seen in many different social media applications, and even in different websites as 'Customer Agents', to engage with those who visit the website and provide them with answers to their queries then and there. One notable example where live chats are used is YouTube.

In live chats (or any chat-based platform, for that matter), the message that is sent the latest is the one that is visible first and foremost, thus behaving just like a stack. Stack operations such as push and peek are very much applicable here.

3) Backtracking Algorithms

				8	9	5		
		1	2					4
7						2	3	
2								
	4	5	1		7	6	8	
								5
	7	8						6
5					3	4		
		9	6	4				

Backtracking can be defined as a general algorithmic technique that considers searching every possible combination in order to solve a computational problem.

It helps to solve problems recursively by trying to build a solution incrementally, one piece at a time, and removing those solutions that fail to satisfy the conditions stated by the problem. So, add solution piece one by one, and remove it if it isn't a plausible solution as per the constraints of the given problem.

This is *exactly* how a stack works.

Let us understand it further using an example of Sudoku.

In the Sudoku problem, we try filling digits one by one in order to achieve the solution, which is having the numbers 1 to 9 in each row, column, and box without any repetitions. With each digit we fill in a particular position, we check if that position is applicable for that digit. If it is, we keep the digit in our *stack of solutions*, and if not, we remove it.

So, in every instance of solving the problem, we are only dealing with the latest element – we either see if the element we have *pushed* is valid, and if it is not, we immediately have to *pop* it.

APPLICATIONS OF QUEUE

1) TV Episodes Release Order

No. overall	No. in season	Title	Directed by	Written by	Original release date
1	1	"It's About Time"	Robert Valley	Robert Kirkman	March 25, 2021
When hulking villainous scientists, the Mauler Twins, attack the White House, they are repelled by the Guardians of the Globe and Nolan Grayson / Omni-Man. Nolan's son Mark, who is waiting for his powers to emerge, goes to school and defends classmate Amber Bennett from bully Todd. After Todd beats Mark, Amber saves the latter and develops an interest in him. Later, Mark's powers emerge, and he tells his parents over dinner. While Nolan is initially uncertain, he nonetheless begins to train Mark on how to use them. However, wishing to emphasize the severity of being a superhero, Mark is caught off-guard by his father punching him too hard. Feeling hurt emotionally and physically, Mark blows off steam by stopping a robbery in a homemade suit. After a heart-to-heart with his son, Nolan takes Mark to meet superhero suit tailor Art Rosenbaum, who creates a proper superhero suit for Mark after he chooses to call himself "invincible". Later, Nolan secretly ambushes the Guardians at their headquarters and kills them all before falling unconscious from damage sustained from their resistance.					
2	2	"Here Goes Nothing"	Paul Furminger	Simon Racioppa	March 25, 2021
The clandestine Global Defense Agency (GDA) nurses a comatose Nolan back to health, but fail to revive the Guardians. Director Cecil Stedman informs Mark and his mother Debbie. As extra-dimensional aliens called the Flaxans attack, Mark helps the Teen Team hold them off. When the Flaxans rapidly age and retreat, Teen Team leader Robot deduces this was due to a time dilation difference between the Flaxans' dimension and Earth. As Mark recognizes Atom Eve as classmate Samantha Eve Wilkins, they share their secret identities and become friends. The Flaxans return with anti-aging technology, but Mark and the Teen Team destroy them, forcing another retreat. The Flaxans return once again and nearly succeed until a recuperated Nolan forces them back to their dimension, where he devastates their planet in retaliation before returning as news breaks of the Guardians' deaths. Meanwhile, Mark skirmishes with Allen the Alien, who wants to test Earth's defenses for the Coalition of Planets. Using a time-out to talk and learning of Allen's mission, Mark helps Allen realize he confused "Earth" for another planet called "Urath" and leaves on good terms. Concurrently, demon detective Damien Darkblood investigates the Guardians' deaths for Cecil, theorizing the killer was among the heroes.					

The episodes of a television series are *generally* released as per their filming/production order. That is, whichever episode is made first is released first. This is how a queue works.

So an episode is filmed, *enqueued* into the release queue, and when it is ready to be released to the audience, it is *dequeued*. We can *peek* into the front of the queue to see the next episode that is the released, which is another queue operation. We can check if the queue is empty, that is, check if there any episodes available to be released. We can check if the queue is full, which means checking whether all the episodes are ready to be released.

Of course, there are discrepancies in real-life. Usually, it might be some scenes from the last episode that are shot first and released first. This is a possibility, yes. Sometimes, the entire last half of a television series is filmed before the first half is filmed. However, mostly, the episode release order adheres to the filming order, and thus the queue logic can indeed be applied here.

2) Vehicles on Toll Gate

Vehicles form a queue on a toll gate. The first vehicle which enters the queue has to pay the toll charges in order to proceed to the highway. Vehicle by vehicle, the first-in first-out policy is in effect here.

3) MIT Registration Number Allotment

Previously, register numbers in MIT were allotted in the alphabetical order, which is a standard that is followed in most schools and colleges. However, in 2022, the system was changed. Register numbers were allotted based on whoever got themselves admitted into the institution first for their respective branches. That is, it now follows the first-come first-serve policy, which is how a queue works. This streamlines the admission process and allows the institution to set up students' register numbers at the first step. Anyone who joins at a later period will simply get the last register number available to them.

2022506065	Guhan M	
2022506066	Shalini k	
2022506067	Mona K	
2022506068	J.vaishnavi	
2022506069	Deepika. M	
2022506070	TAMILSELVAN	
2022506071	Malik Abdur Rahman J	
2022506072	Aashin A P	
2022506073	Sushmidha B	
2022506074	Vamika V	
2022506075	T anushri	
2022506076	Ajay R	
2022506077	Niyathi SB	
2022506077	S B niyathi	
2022506078	E.Kabil3sh Selvakumar	
2022506079	Yuvaraj K	
2022506080	KALPANA S	
2022506081	Manobala P	
2022506082	Barathbala B	
2022506083	Nelson Bharathi A	
2022506084	Gowtham R	

CONVERSION OF INFIX EXPRESSION TO POSTFIX EXPRESSION USING STACK

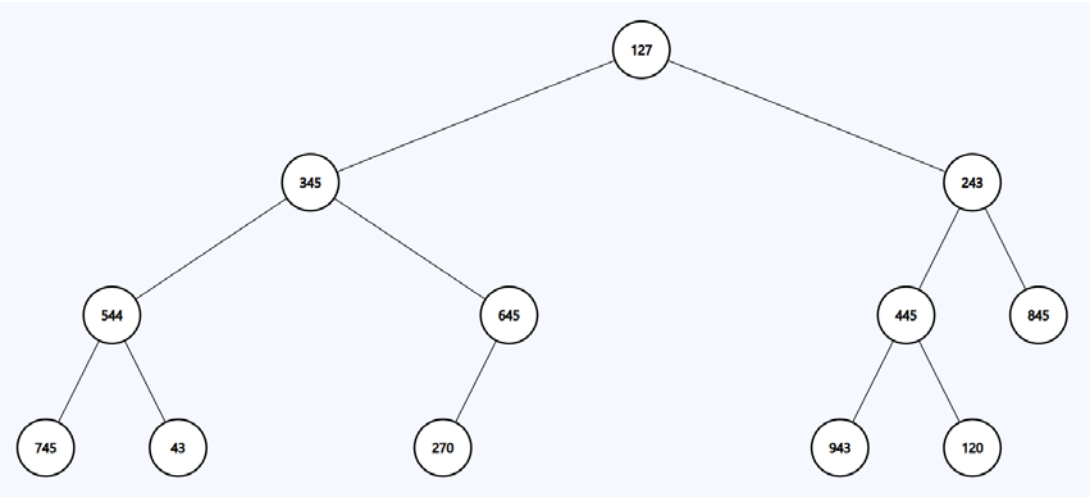
Infix Expression: (A+B)-30*(C^(2-D/E)+F)*(G*40)/5-H*(I-J)

Current Symbol	Operator Stack	Postfix String
(	(	
A	(	A
+	(+	A
B	(+	AB
)		AB+
-	-	AB+
30	-	AB+30
*	-*	AB+30
(	-*(	AB+30
C	-*(	AB+30C
^	-*(^	AB+30C
(	-*(^(	AB+30C
2	-*(^(	AB+30C2
-	-*(^(-	AB+30C2
D	-*(^(-	AB+30C2D
/	-*(^(-/	AB+30C2D
E	-*(^(-/	AB+30C2DE
)	-*(^	AB+30C2DE/-
+	-*(+	AB+30C2D/-^
F	-*(+	AB+30C2D/-^F
)	-*	AB+30C2D/-^F+
*	-*	AB+30C2D/-^F*
(	-*(	AB+30C2D/-^F
G	-*(	AB+30C2D/-^FG
*	-*(*	AB+30C2D/-^FG
40	-*(*	AB+30C2D/-^FG40

)	-*	AB+30C2D/-^FG40*
/	-/	AB+30C2D/-^FG40**
5	-/	AB+30C2D/-^FG40**5
-	-	AB+30C2D/-^FG40**5/-
H	-	AB+30C2D/-^FG40**5/-H
*	-*	AB+30C2D/-^FG40**5/-H
(	-*(	AB+30C2D/-^FG40**5/-H
I	-*(	AB+30C2D/-^FG40**5/-HI
-	-*(-	AB+30C2D/-^FG40**5/-HI
J	-*(-	AB+30C2D/-^FG40**5/-HIJ
)	-*	AB+30C2D/-^FG40**5/-HIJ-

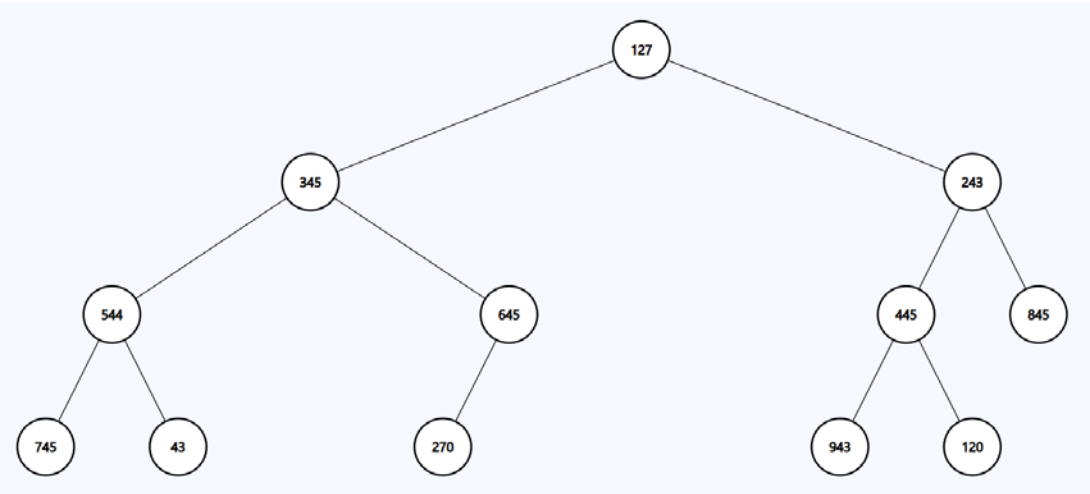
Postfix Expression: AB+30C2D/-^FG40**5/-HIJ-*-

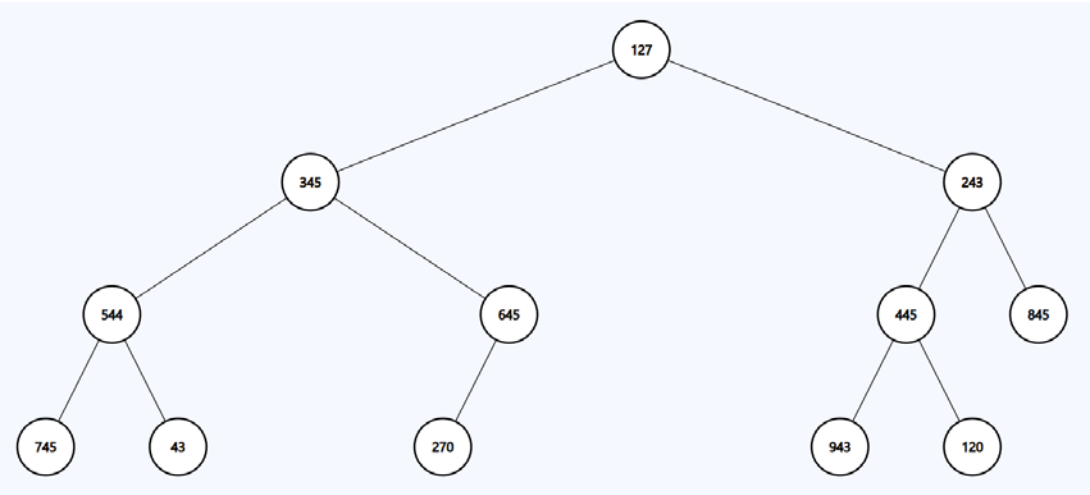
BINARY TREE TRAVERSAL

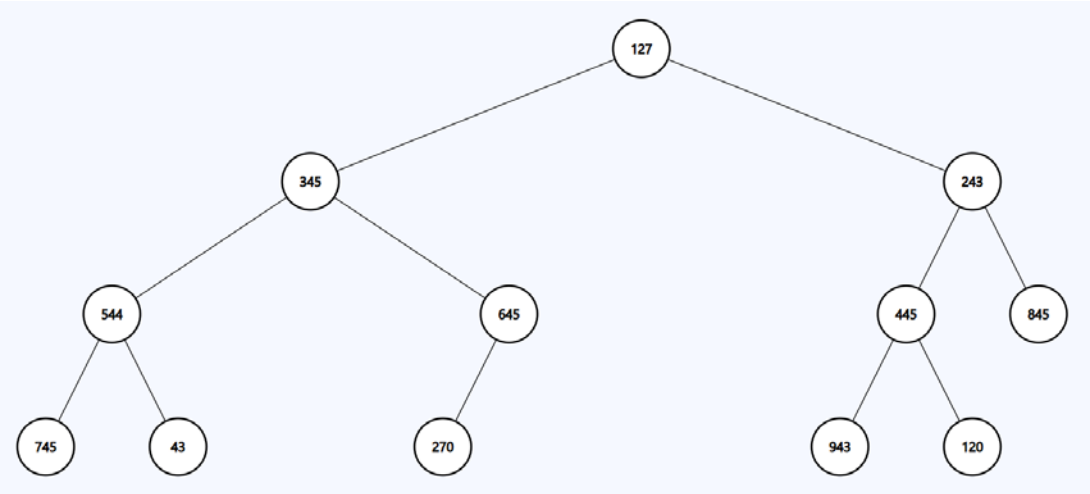


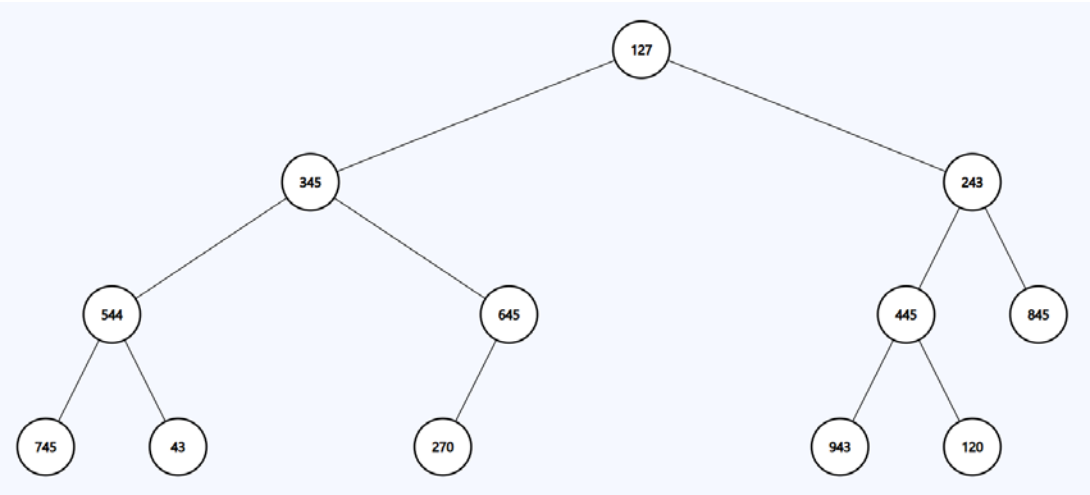
Height of the Binary Tree $h = 3$
Maximum Number of Possible Nodes = $2^{h+1}-1 = 2^4-1 = 15$
Actual Number of Nodes Present = 12

Inorder Traversal – Left, Root, Right

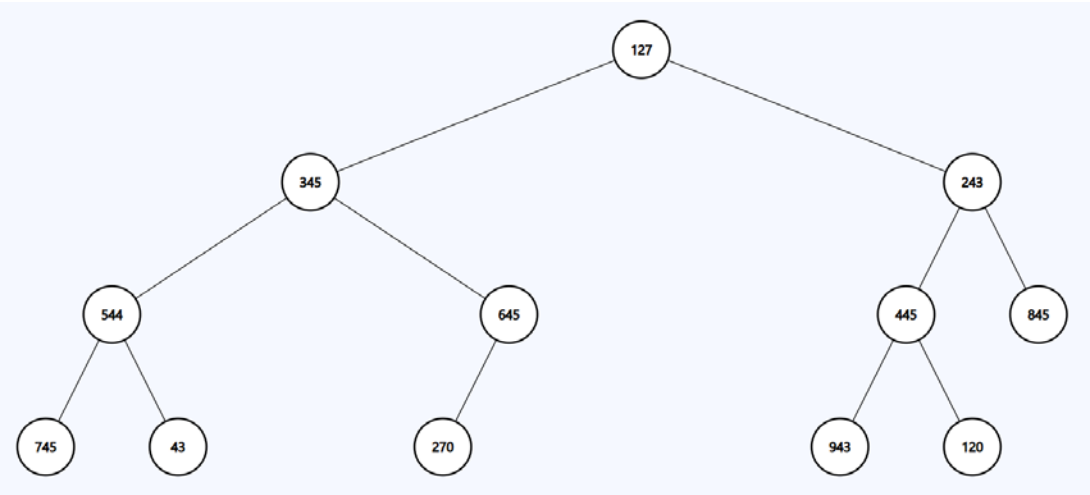




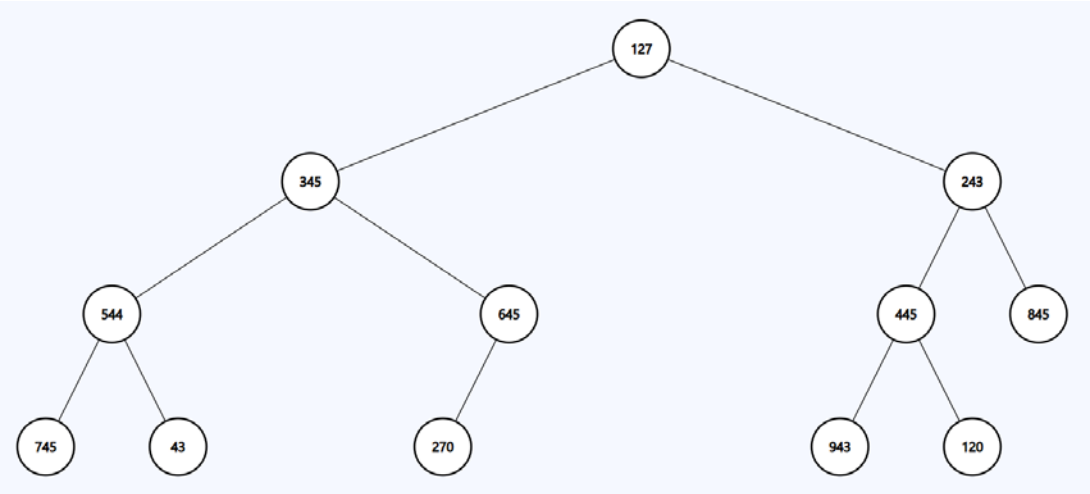




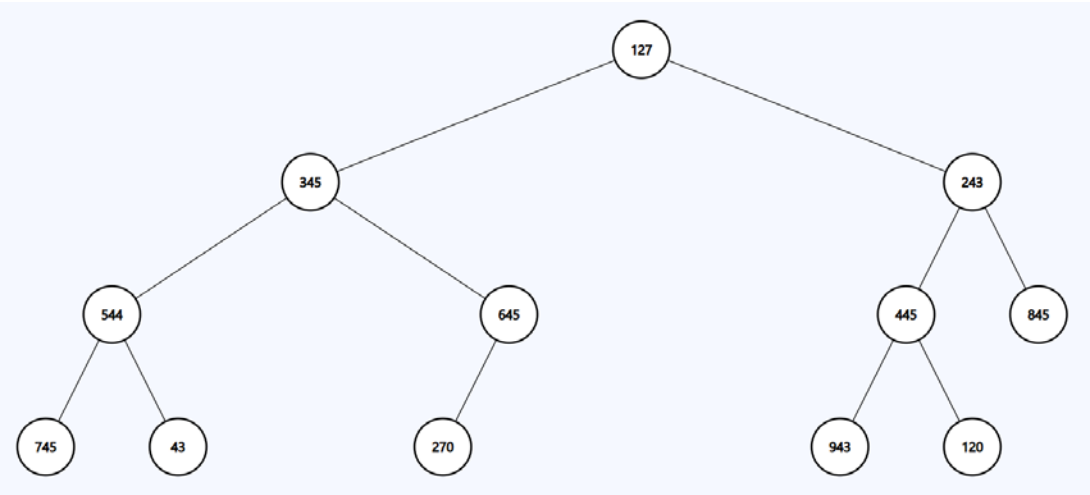
Inorder Traversal: 745



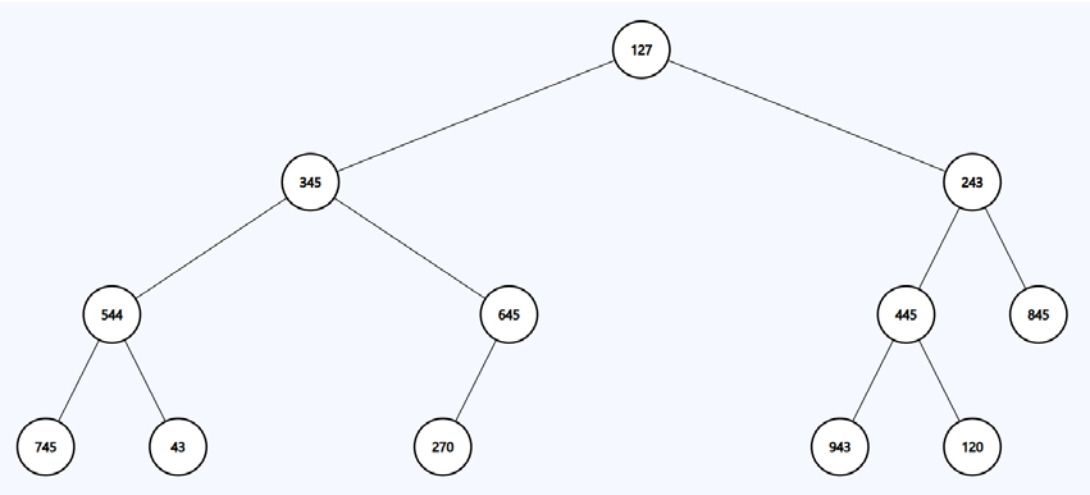
Inorder Traversal: 745, 544



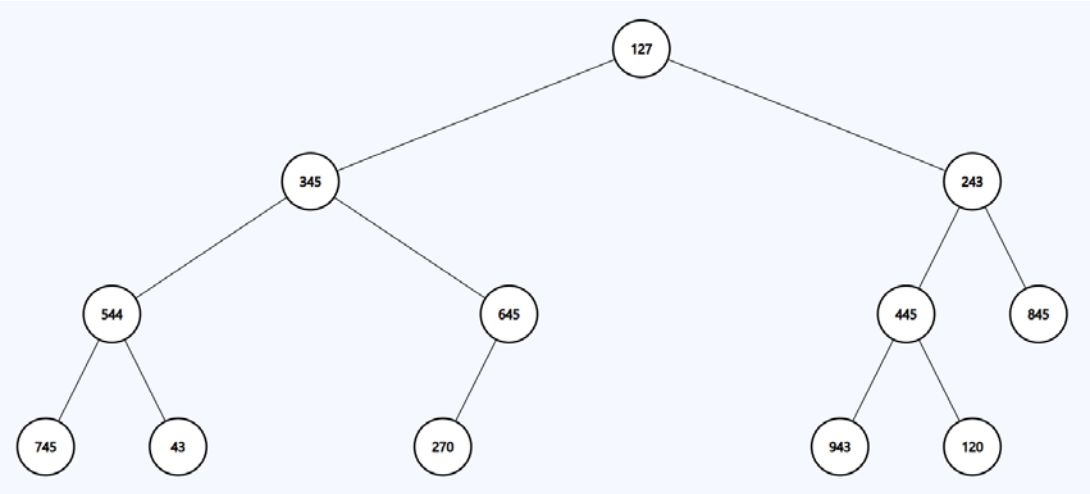
Inorder Traversal: 745, 544, 43



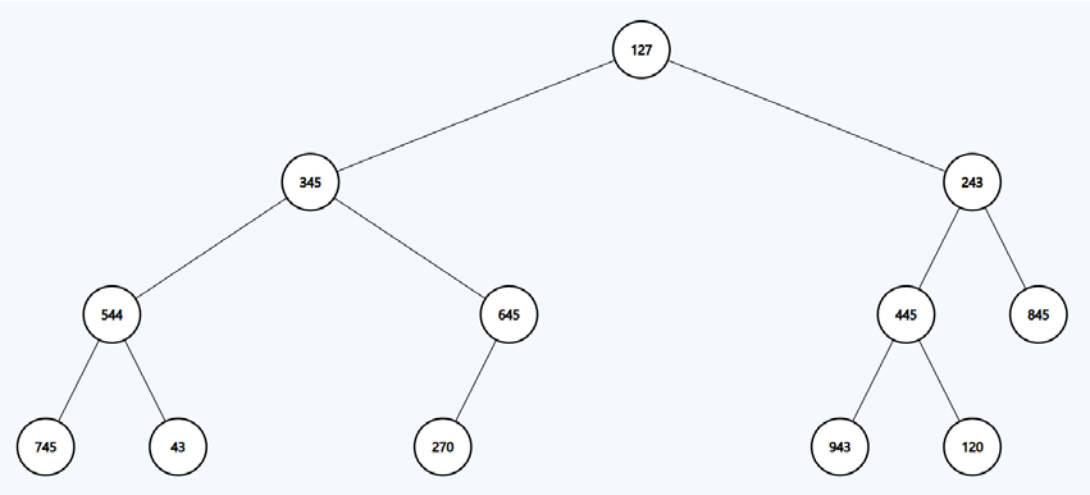
Inorder Traversal: 745, 544, 43, 345



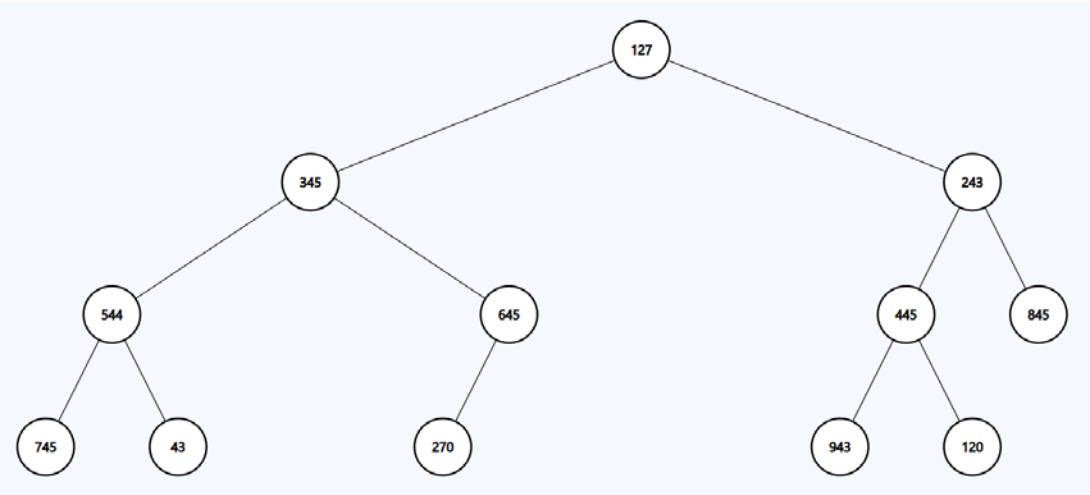
Inorder Traversal: 745, 544, 43, 345



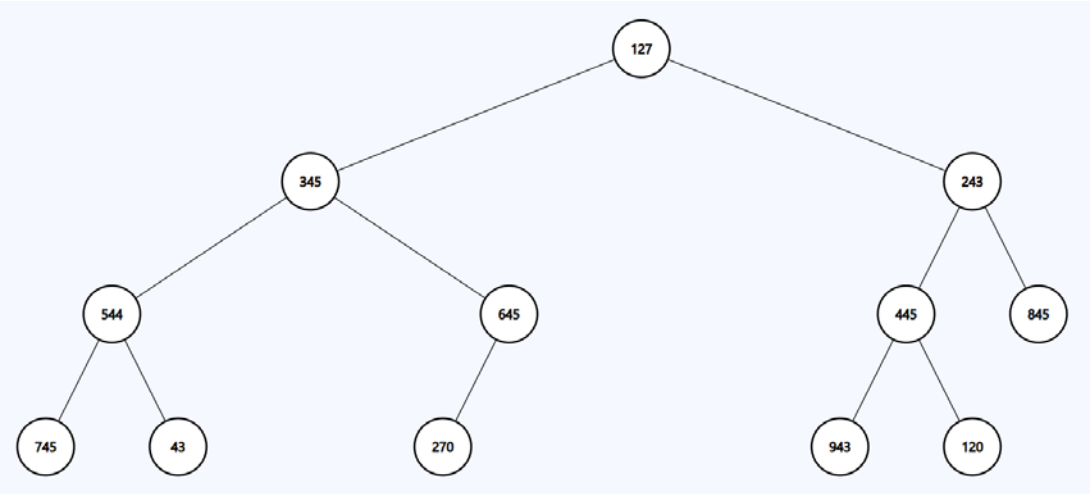
Inorder Traversal: 745, 544, 43, 345, 270



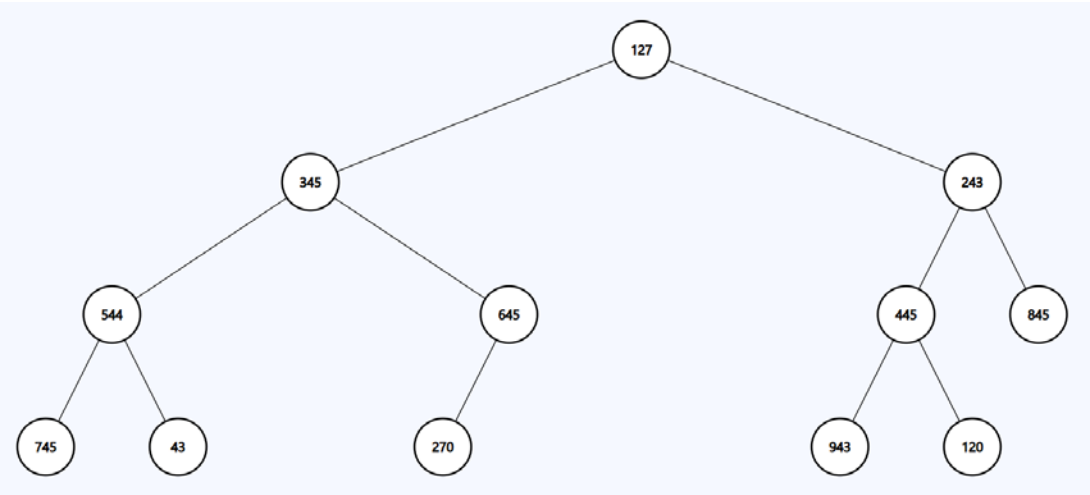
Inorder Traversal: 745, 544, 43, 345, 270, 645



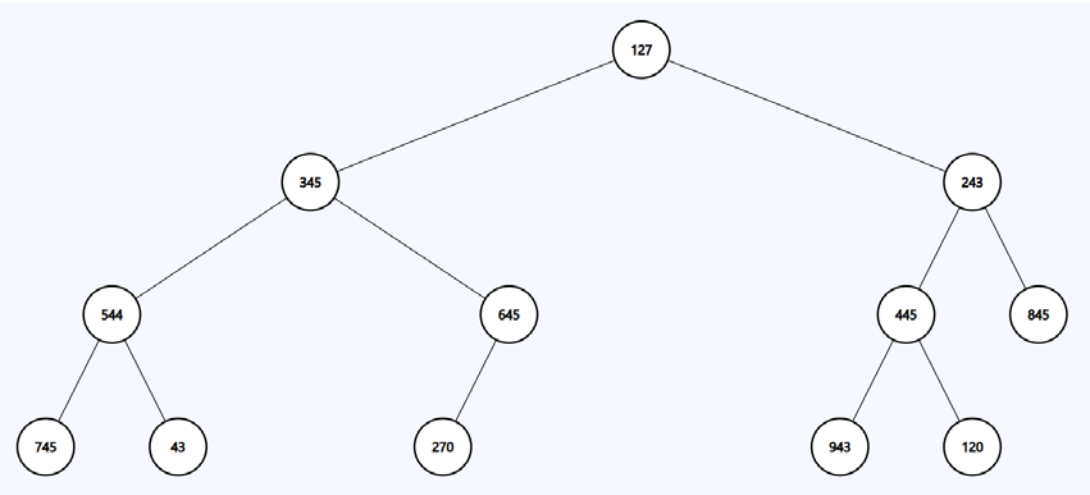
Inorder Traversal: 745, 544, 43, 345, 270, 645, 127



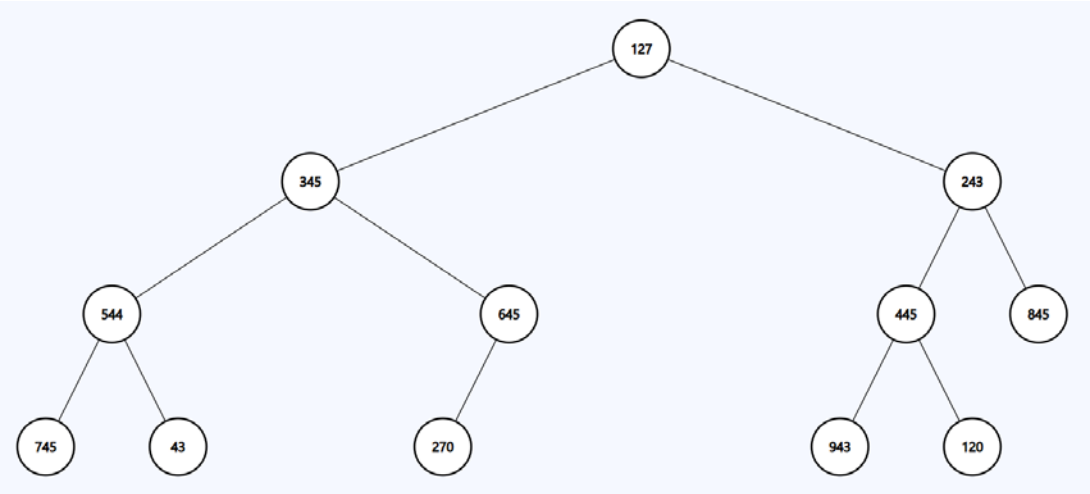
Inorder Traversal: 745, 544, 43, 345, 270, 645, 127



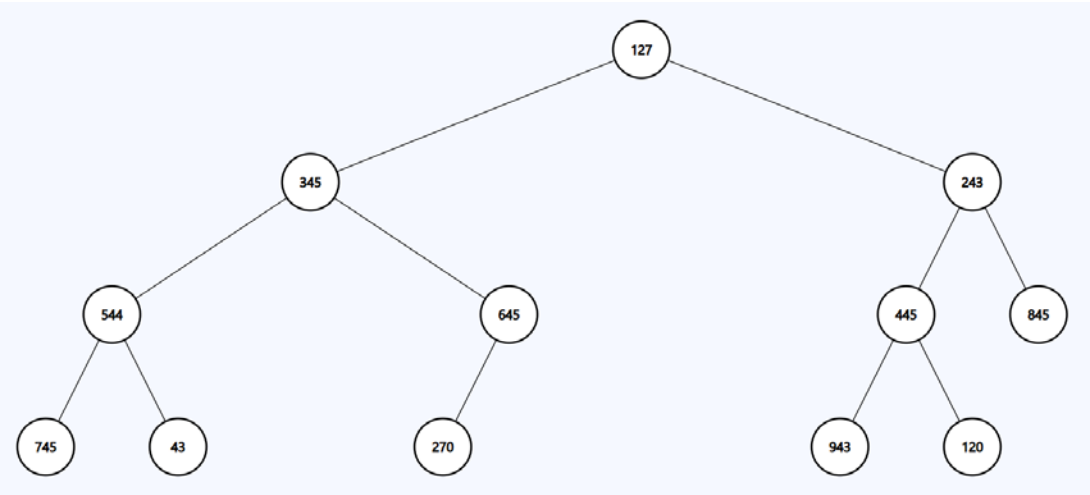
Inorder Traversal: 745, 544, 43, 345, 270, 645, 127



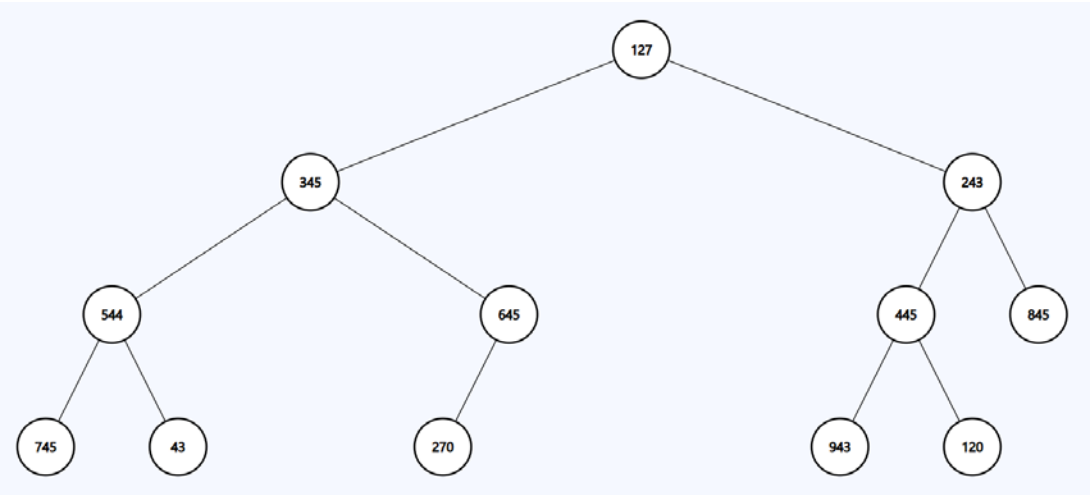
Inorder Traversal: 745, 544, 43, 345, 270, 645, 127, 943



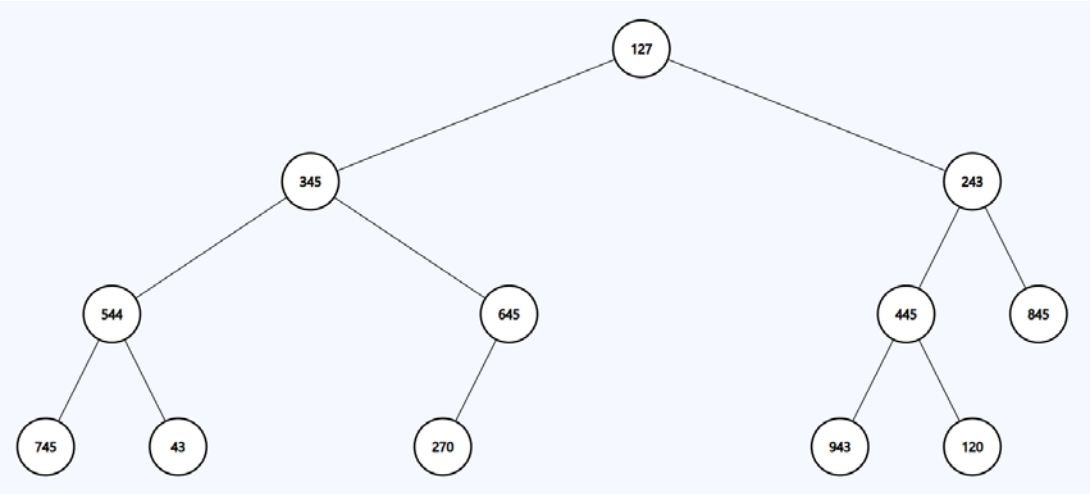
Inorder Traversal: 745, 544, 43, 345, 270, 645, 127, 943, 445



Inorder Traversal: 745, 544, 43, 345, 270, 645, 127, 943, 445, 120



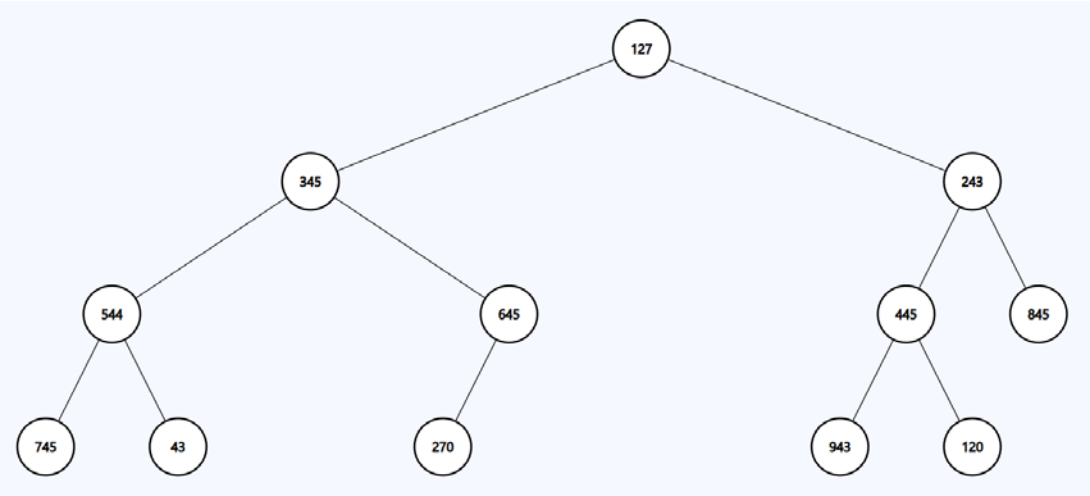
Inorder Traversal: 745, 544, 43, 345, 270, 645, 127, 943, 445, 120, 243



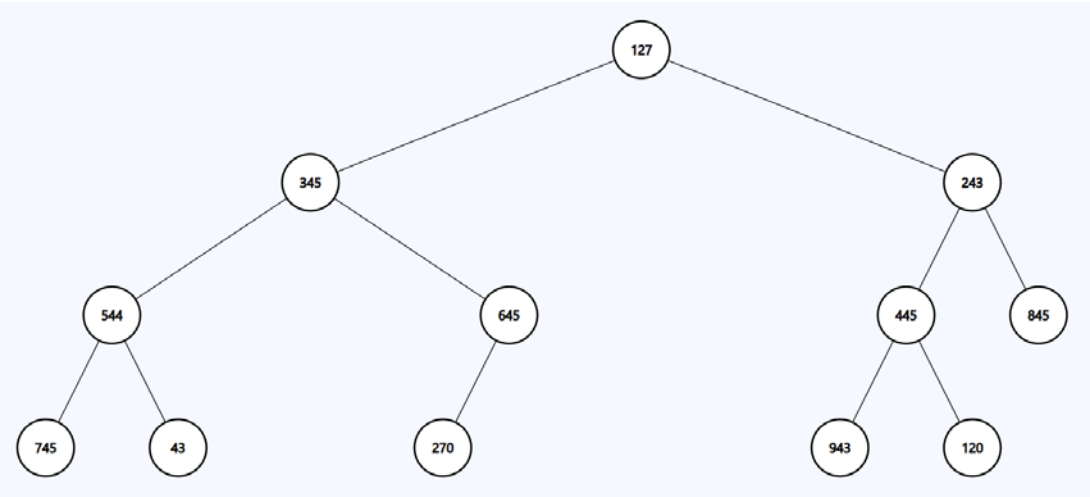
Inorder Traversal: 745, 544, 43, 345, 270, 645, 127, 943, 445, 120, 645

Traversal Complete.

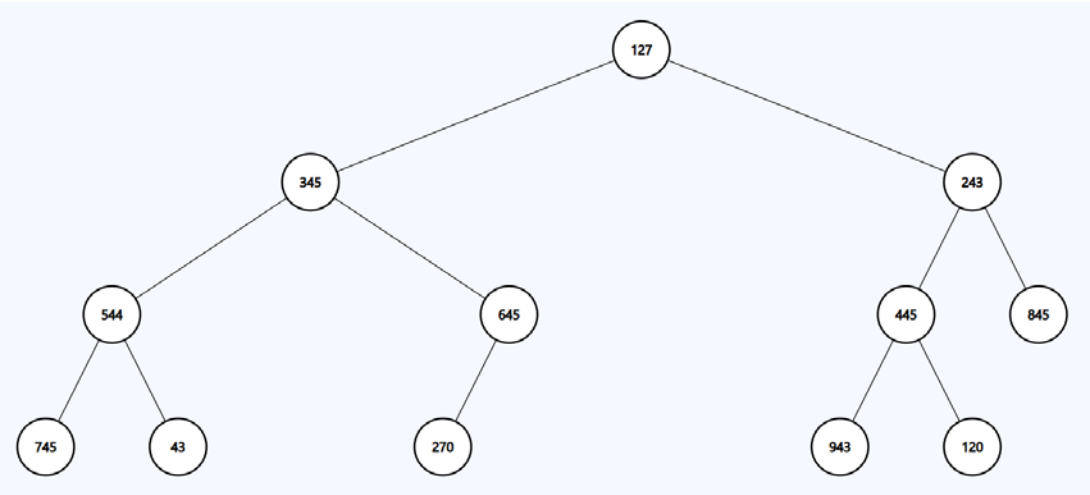
Preorder Traversal – Root, Left, Right



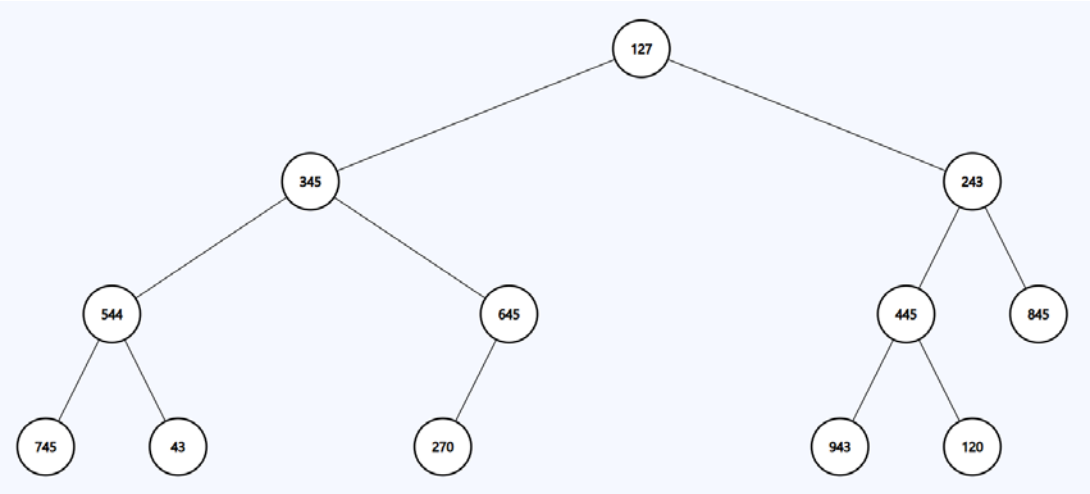
Preorder Traversal: 127



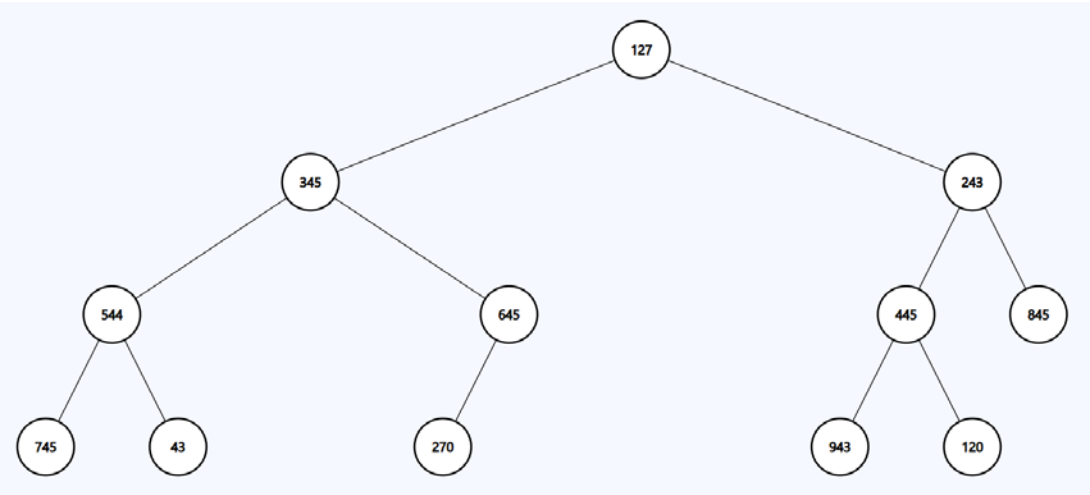
Preorder Traversal: 127, 345



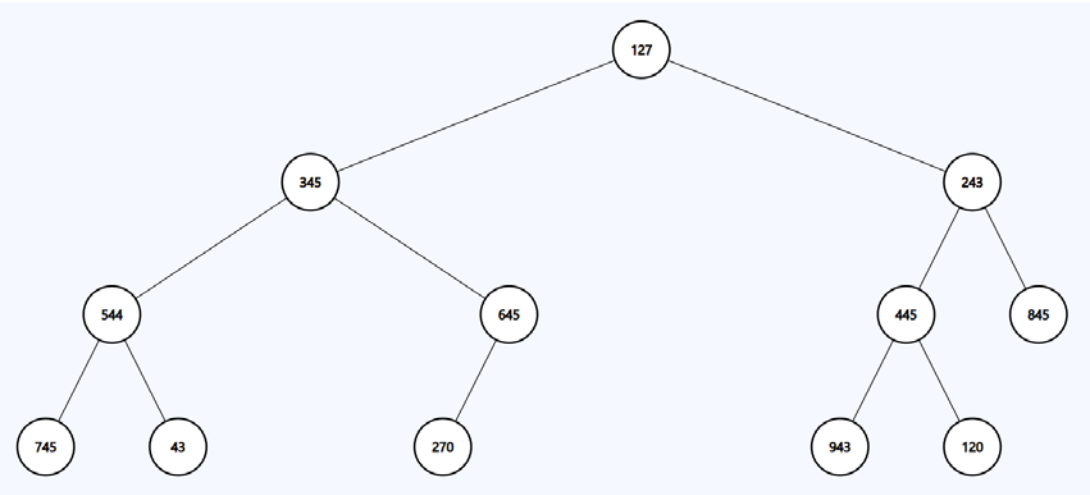
Preorder Traversal: 127, 345, 544



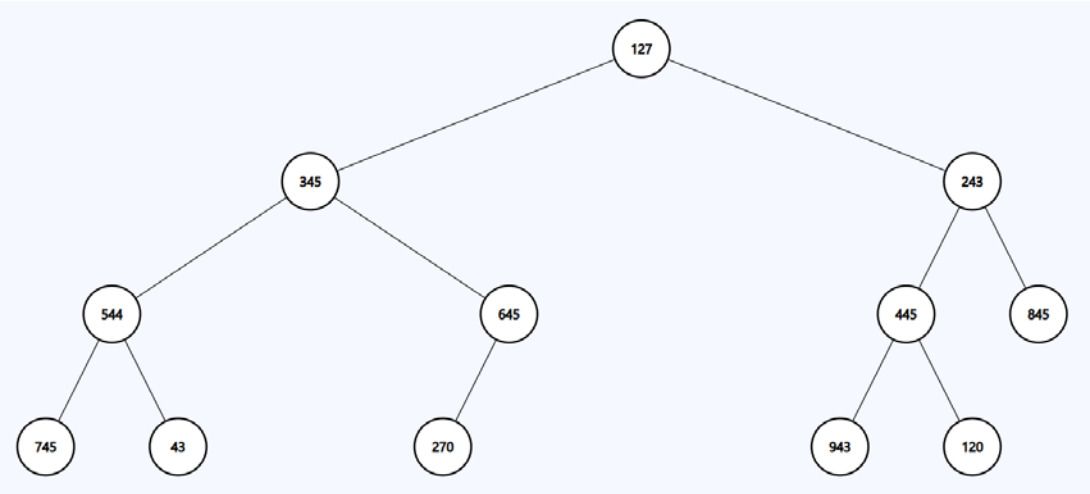
Preorder Traversal: 127, 345, 544, 745



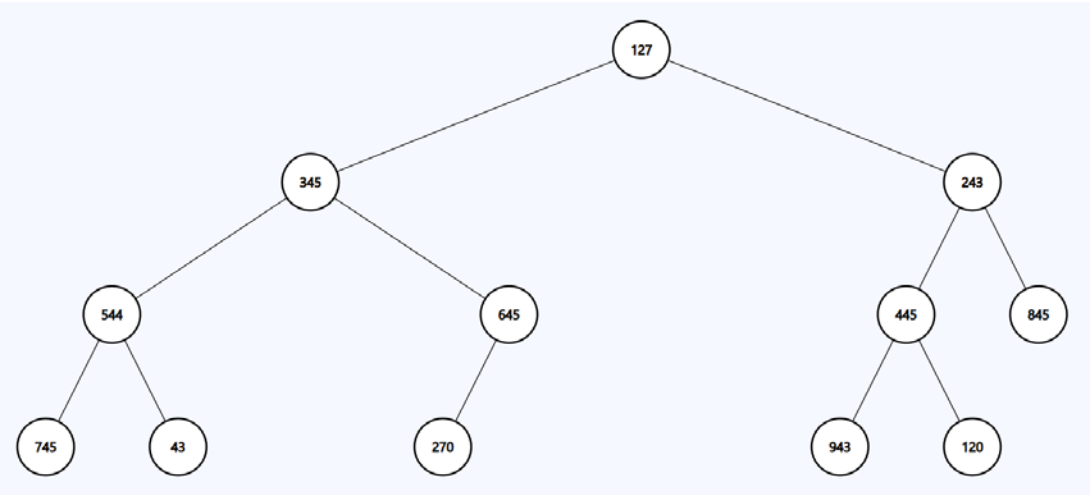
Preorder Traversal: 127, 345, 544, 745, 43



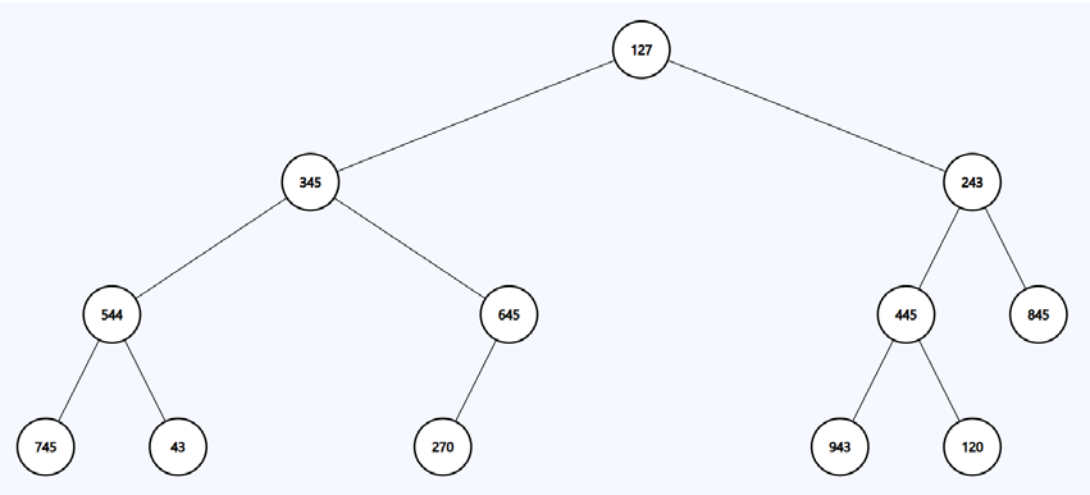
Preorder Traversal: 127, 345, 544, 745, 43, 645



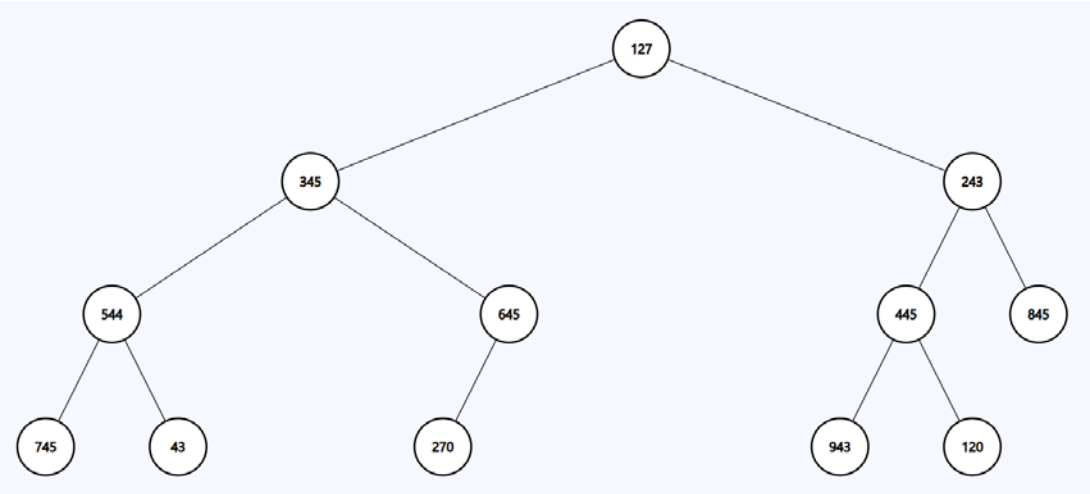
Preorder Traversal: 127, 345, 544, 745, 43, 645, 270



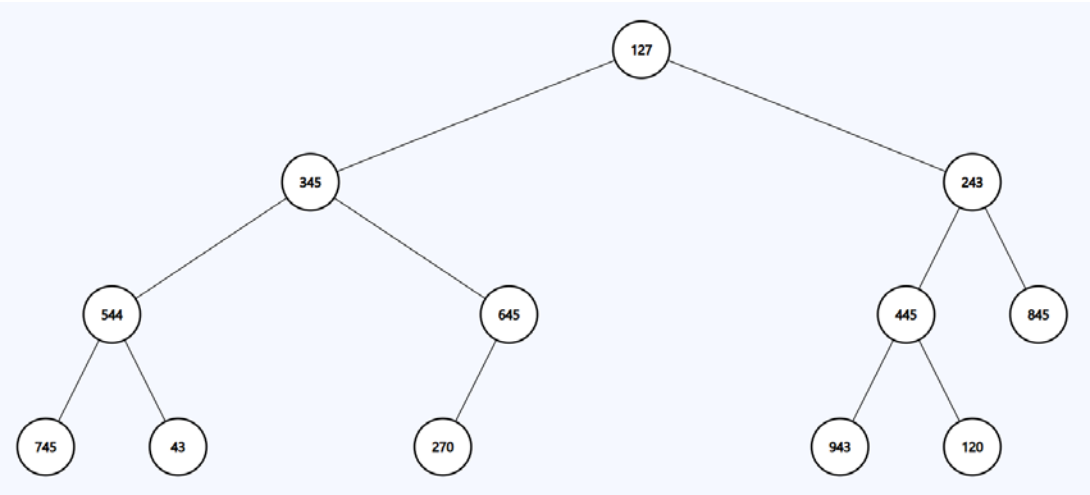
Preorder Traversal: 127, 345, 544, 745, 43, 645, 270, 243



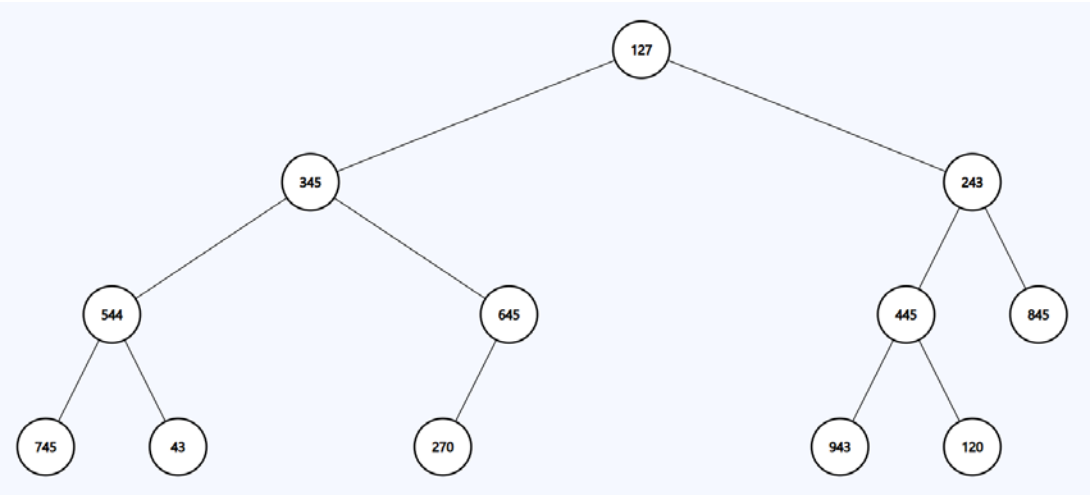
Preorder Traversal: 127, 345, 544, 745, 43, 645, 270, 243, 445



Preorder Traversal: 127, 345, 544, 745, 43, 645, 270, 243, 445, 943



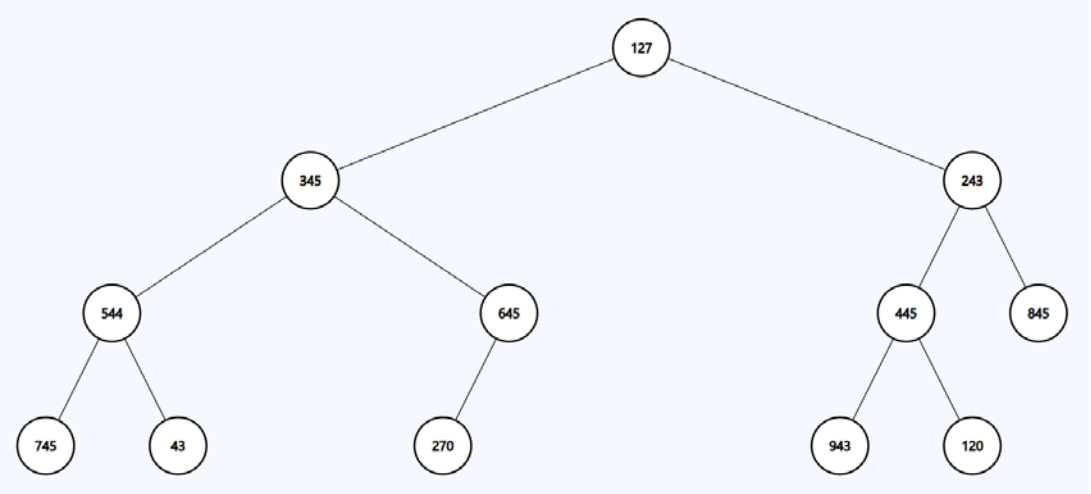
Preorder Traversal: 127, 345, 544, 745, 43, 645, 270, 243, 445, 943, 120

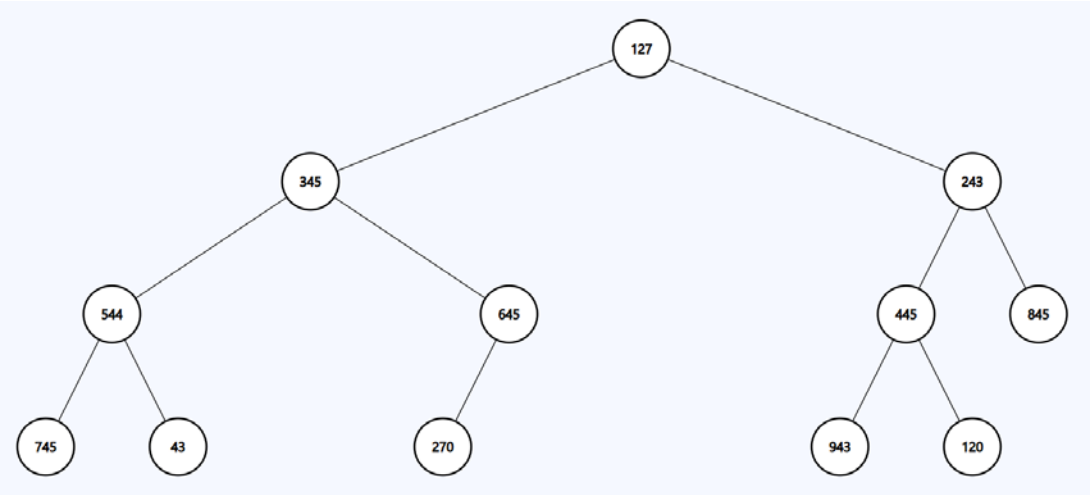


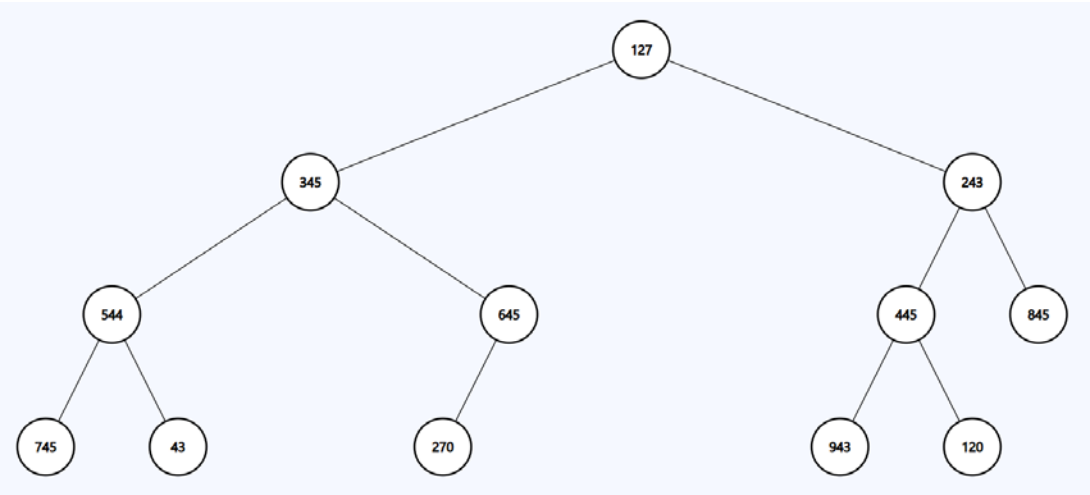
Preorder Traversal: 127, 345, 544, 745, 43, 645, 270, 243, 445, 943, 120, 845

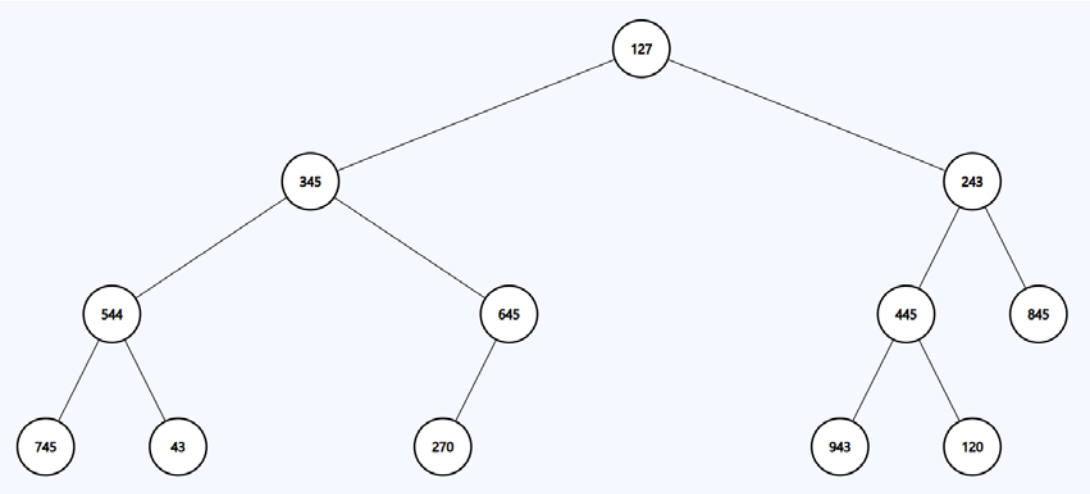
Traversal Complete.

Postorder Traversal – Left, Right, Root

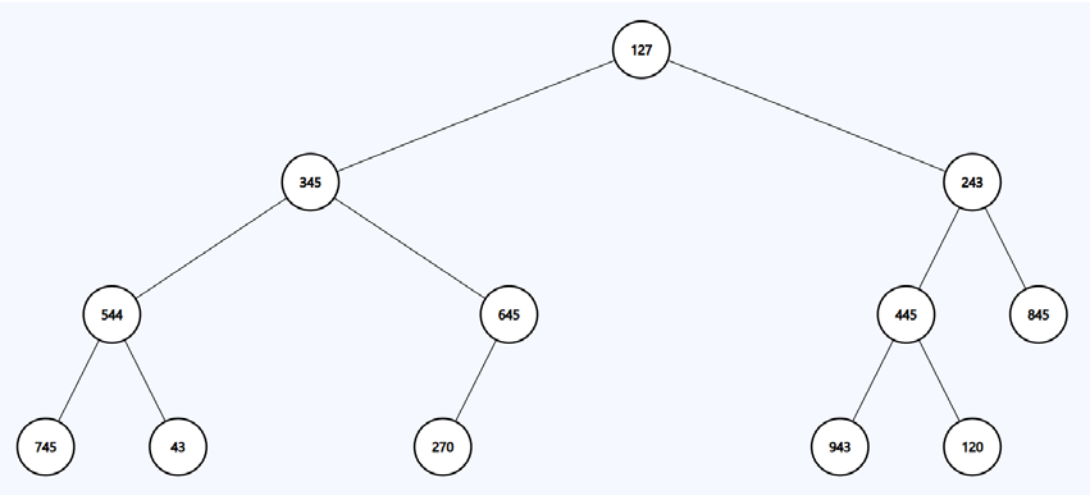




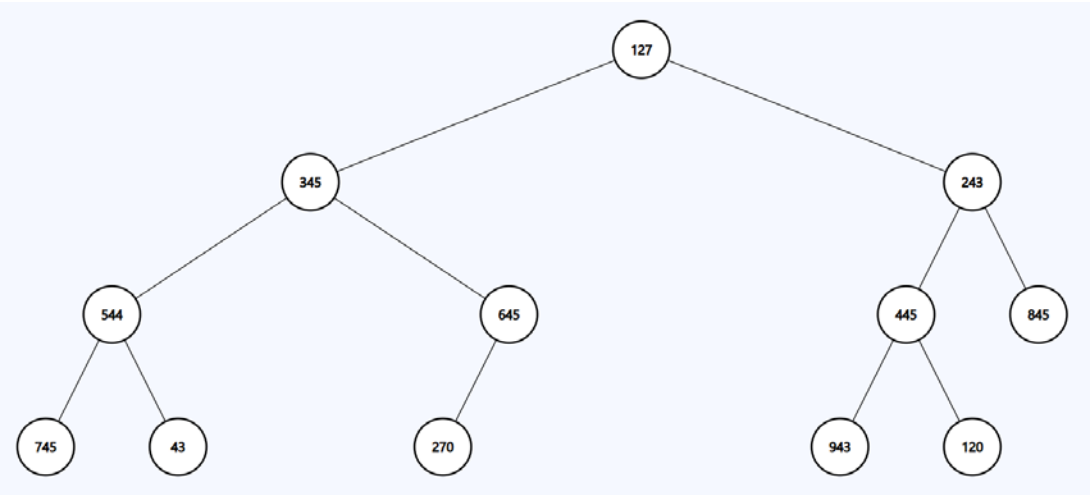




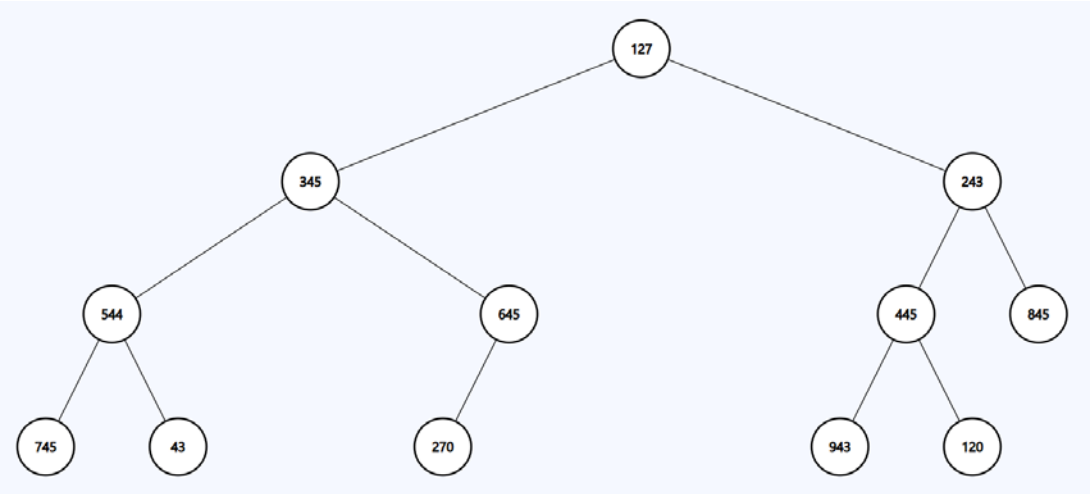
Postorder Traversal: 745



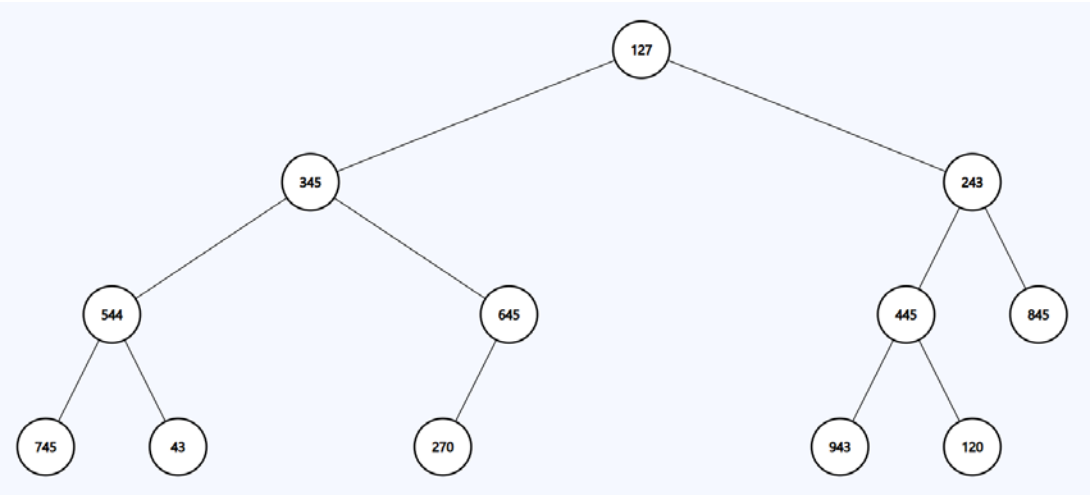
Postorder Traversal: 745, 43



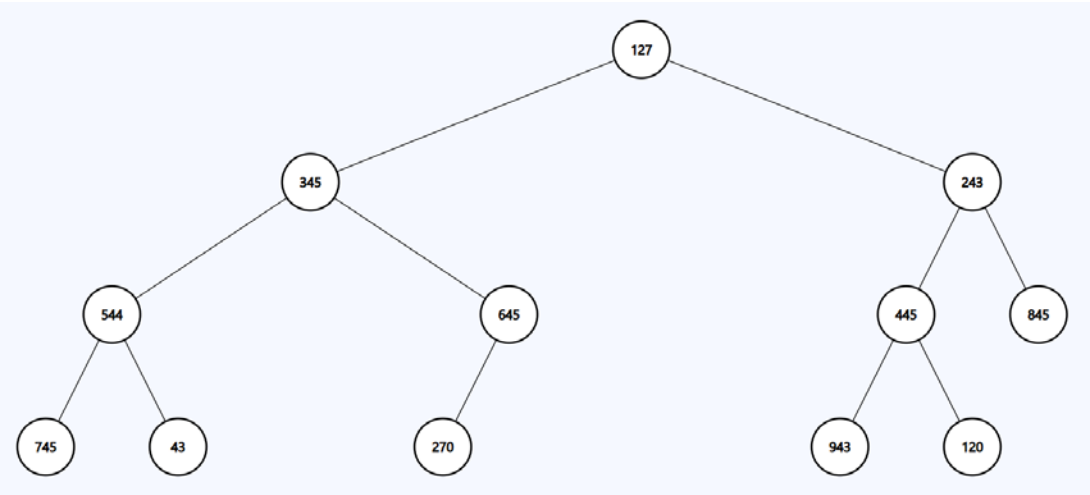
Postorder Traversal: 745, 43, 544



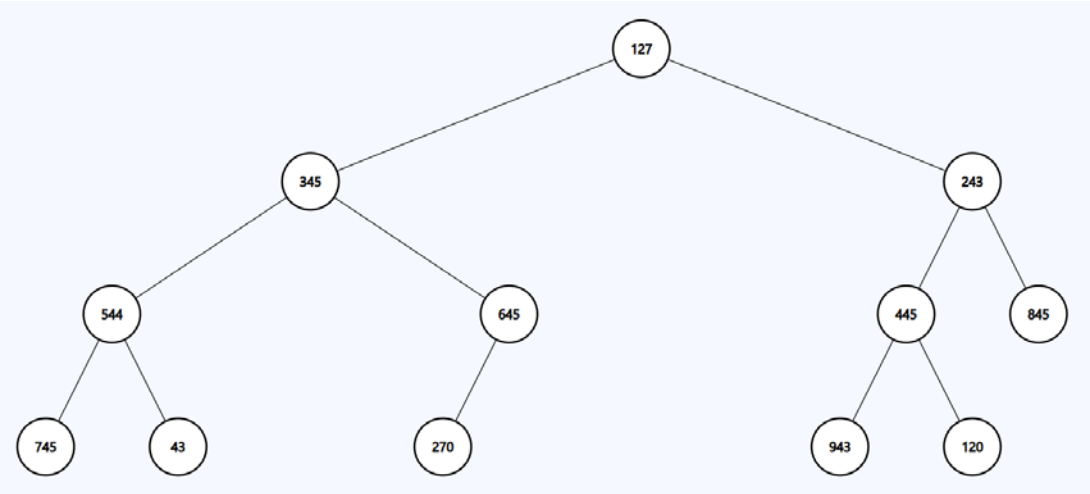
Postorder Traversal: 745, 43, 544



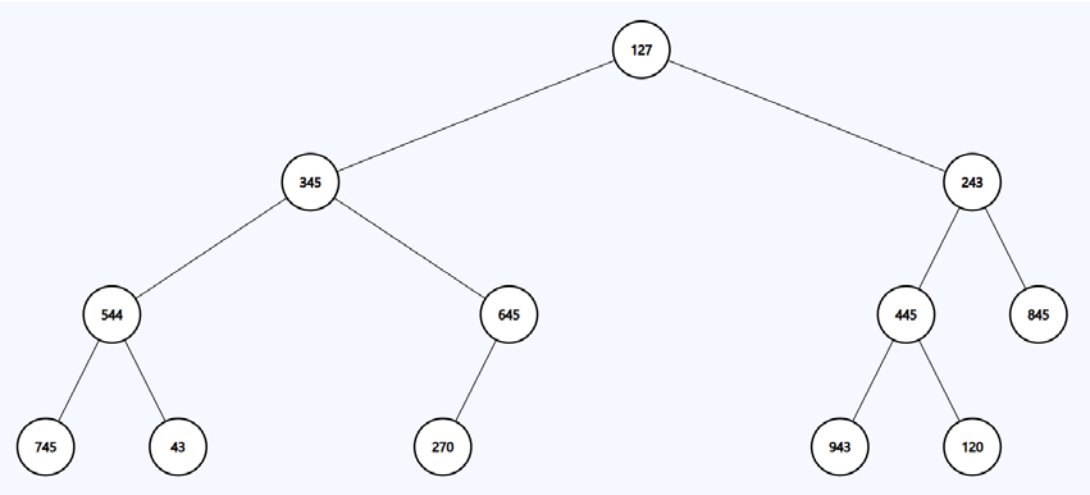
Postorder Traversal: 745, 43, 544, 270



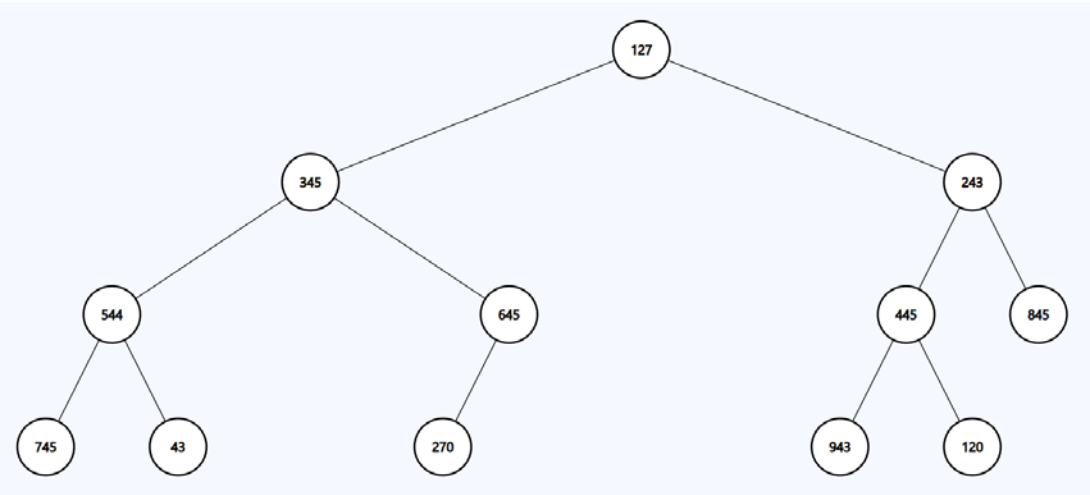
Postorder Traversal: 745, 43, 544, 270, 645



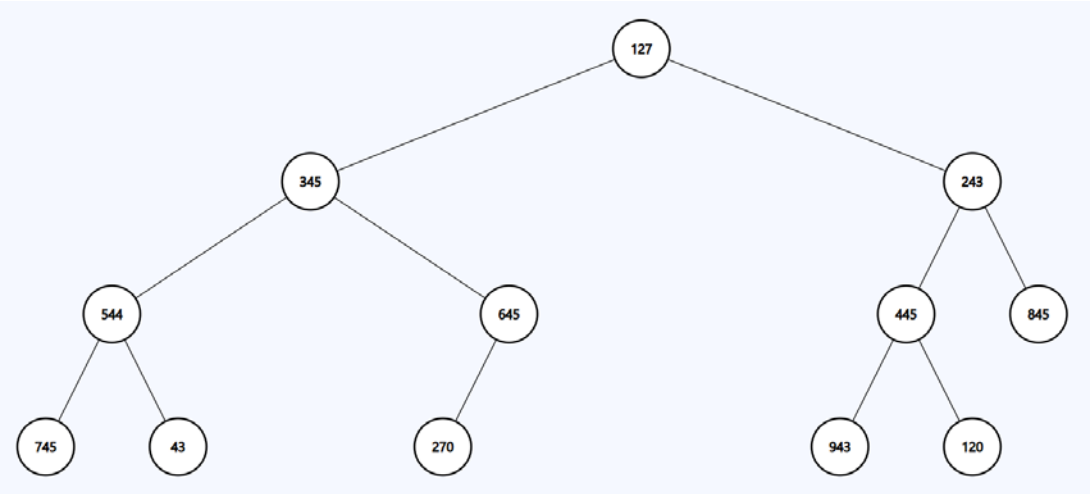
Postorder Traversal: 745, 43, 544, 270, 645, 345



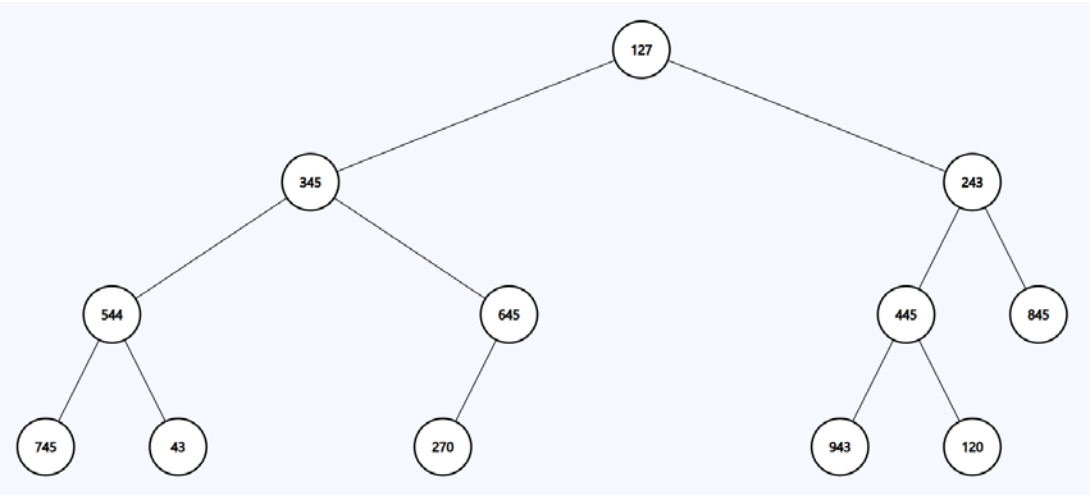
Postorder Traversal: 745, 43, 544, 270, 645, 345



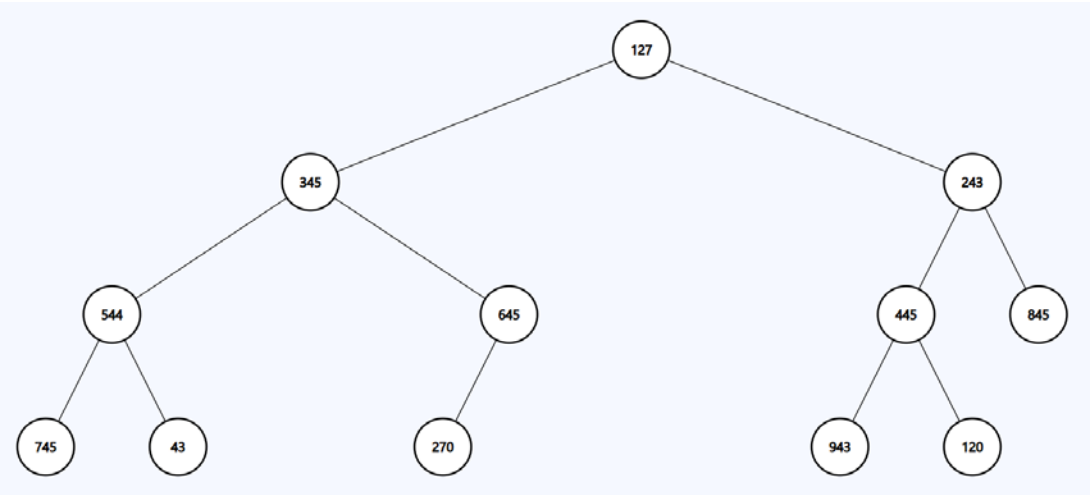
Postorder Traversal: 745, 43, 544, 270, 645, 345



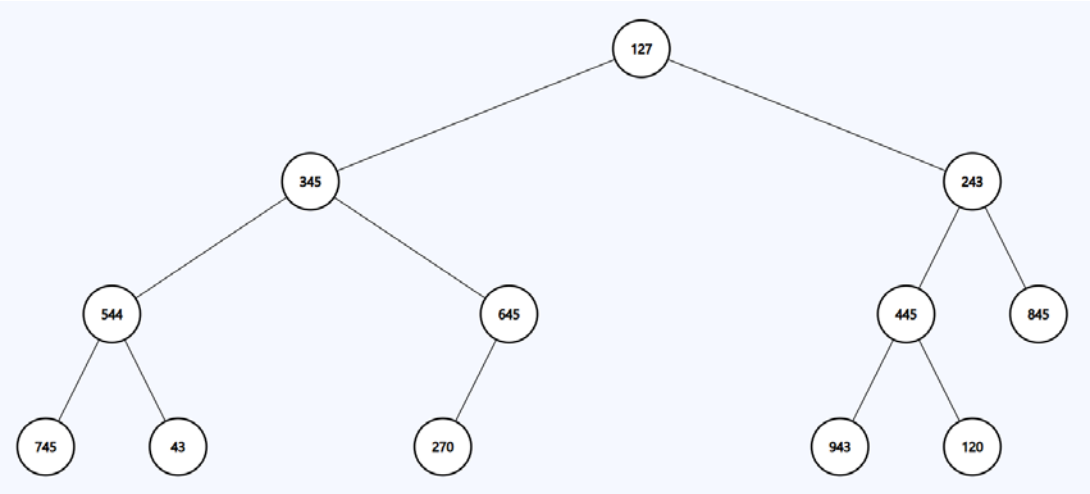
Postorder Traversal: 745, 43, 544, 270, 645, 345, 943



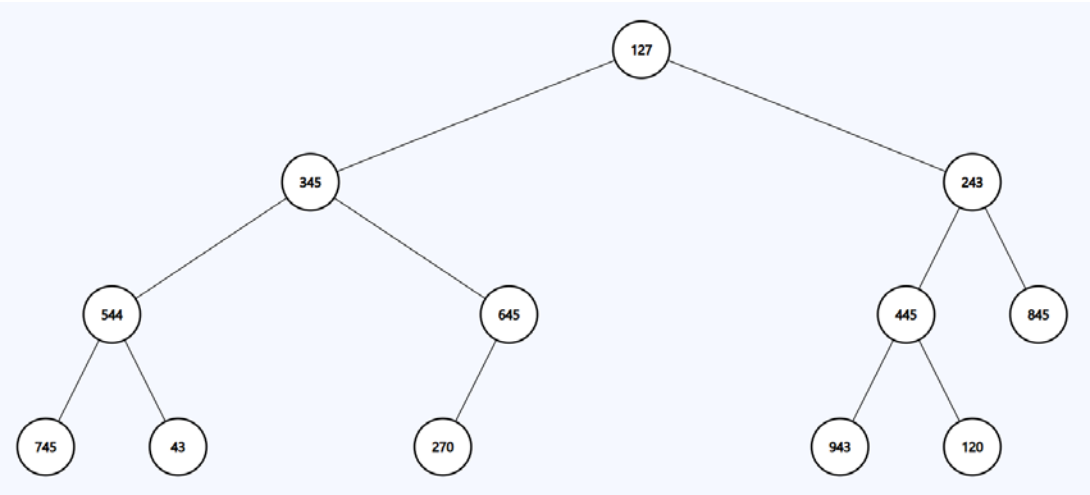
Postorder Traversal: 745, 43, 544, 270, 645, 345, 943, 120



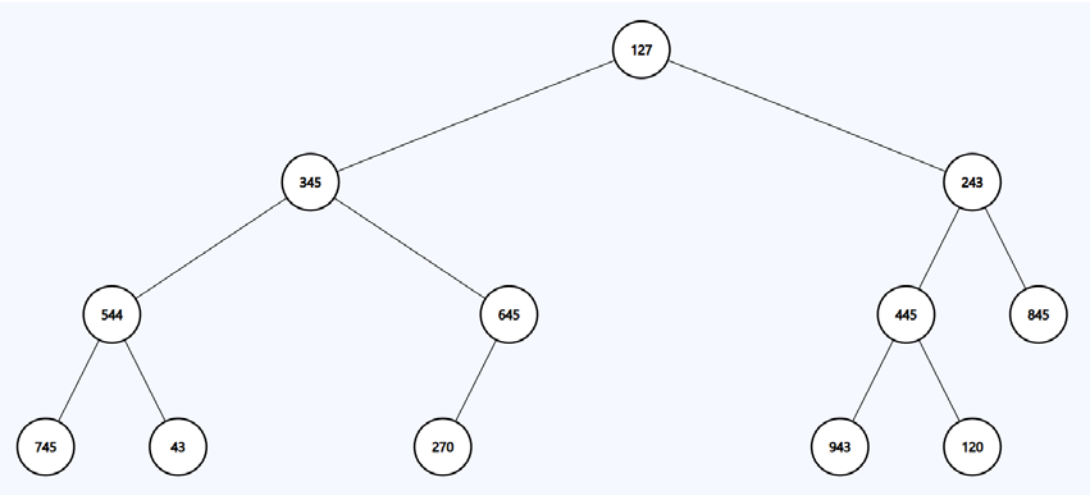
Postorder Traversal: 745, 43, 544, 270, 645, 345, 943, 120, 445



Postorder Traversal: 745, 43, 544, 270, 645, 345, 943, 120, 445, 845



Postorder Traversal: 745, 43, 544, 270, 645, 345, 943, 120, 445, 845, 243



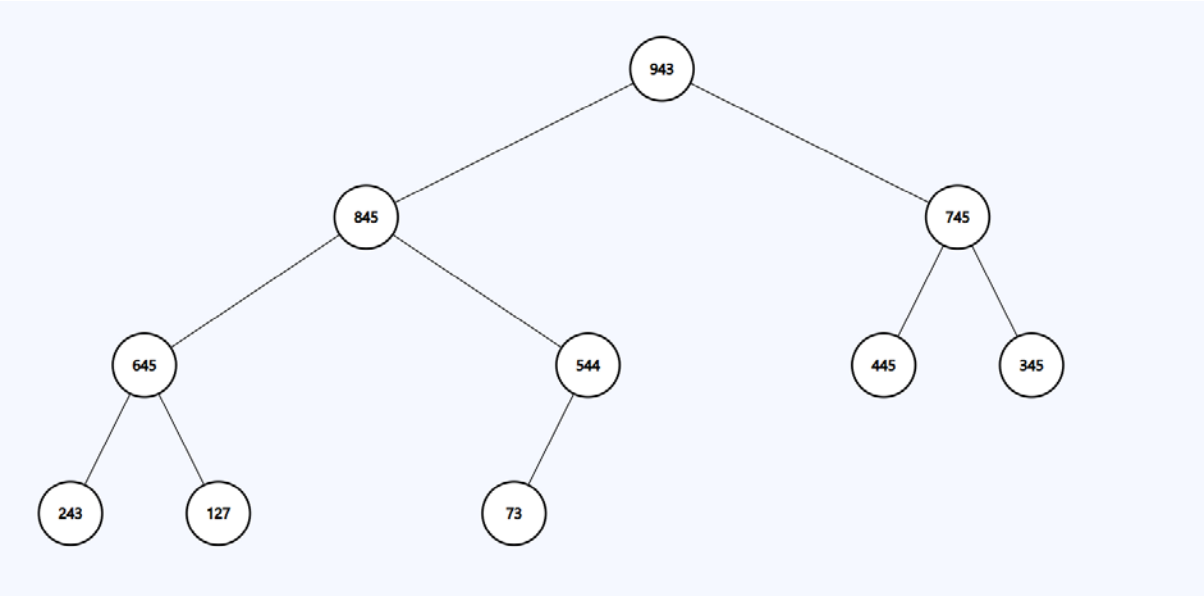
Postorder Traversal: 745, 43, 544, 270, 645, 345, 943, 120, 445, 845, 127

Traversal Complete.

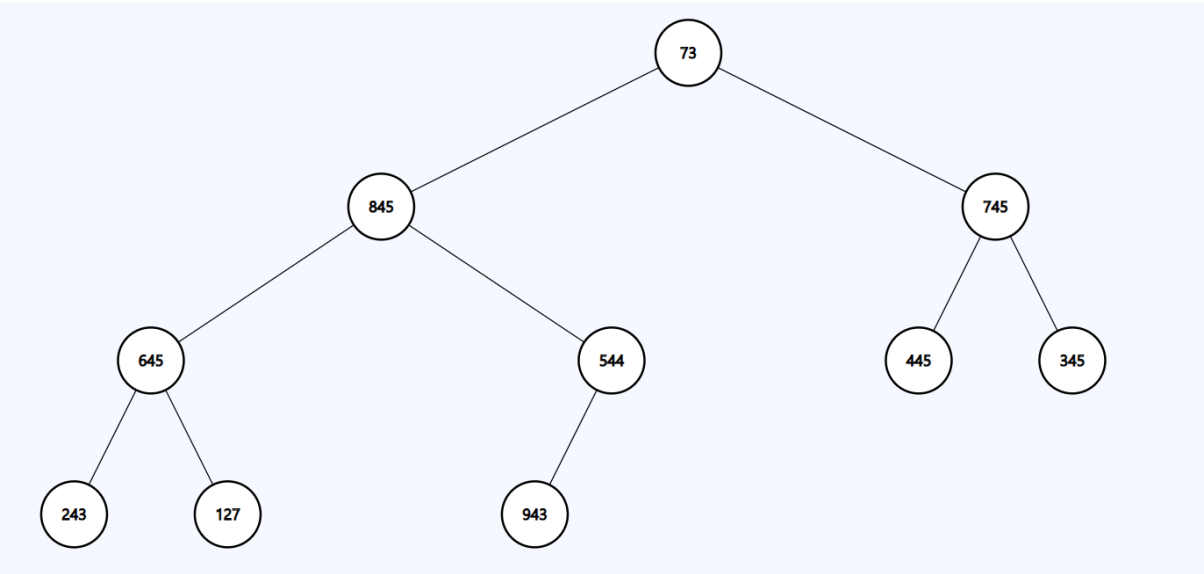
SORTING OF 10 NUMBERS USING HEAP SORT

List of Numbers: 943, 845, 745, 645, 544, 445, 345, 243, 127, 73

Inserting the numbers in a complete binary tree, we get

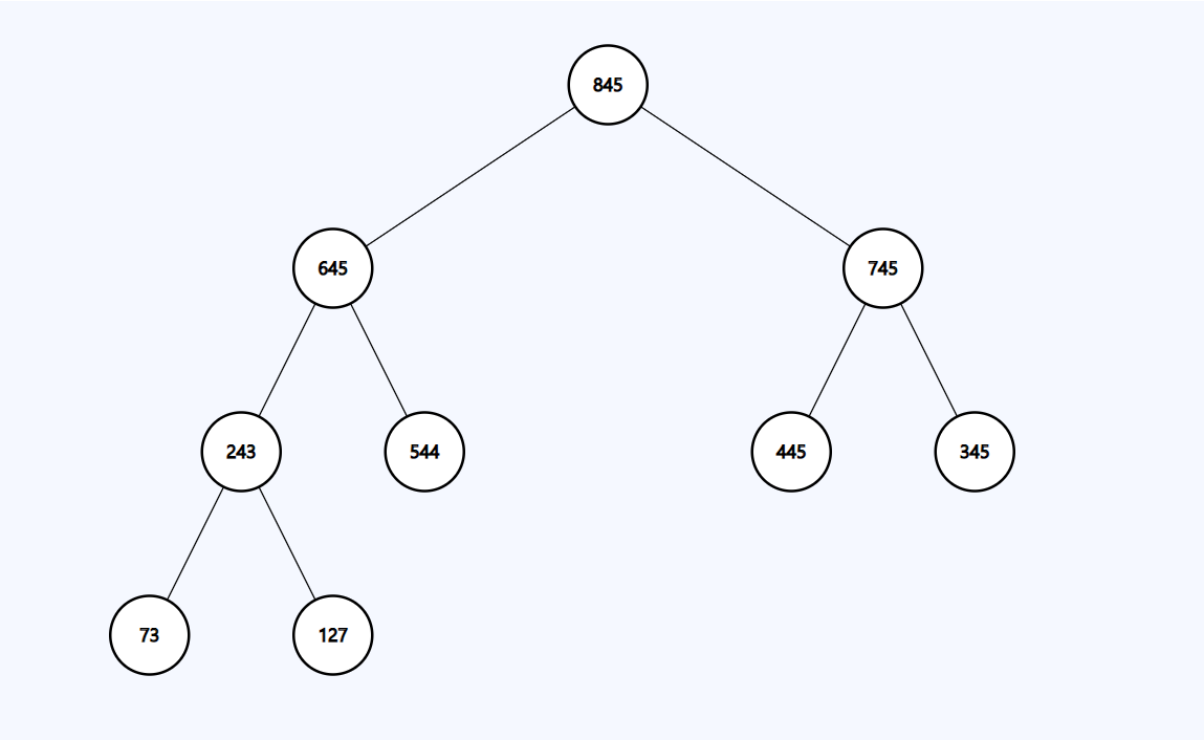


This is already a max heap. Let us perform Heap Sort. Swap 943 and 73, then remove 943.

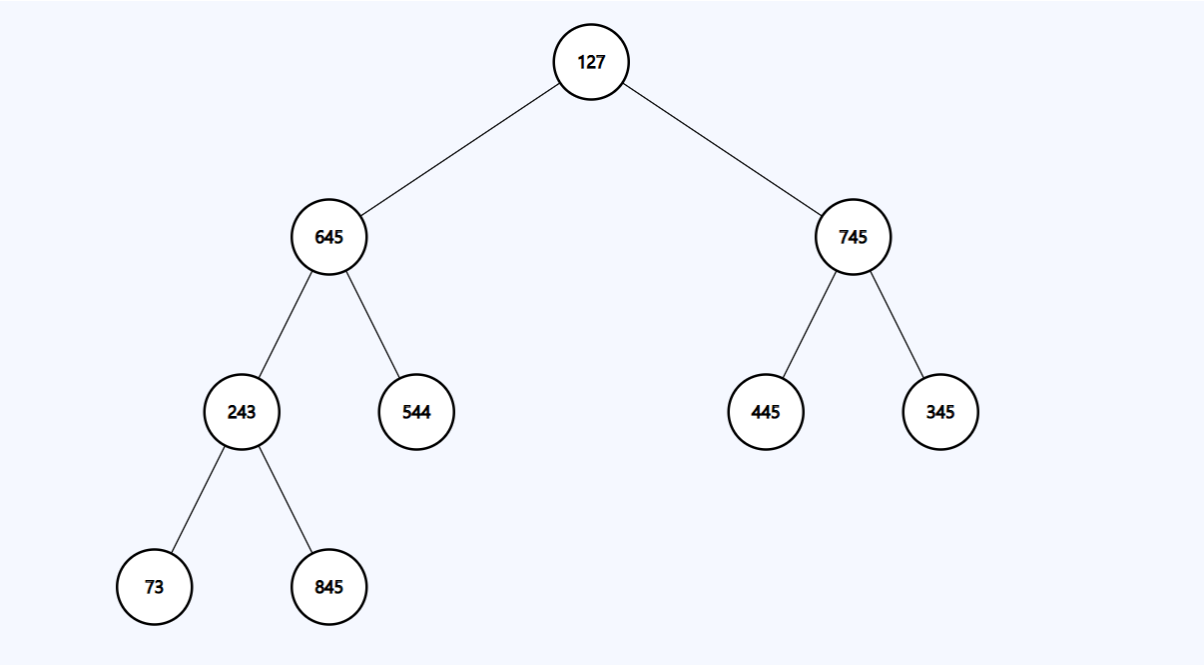


Sorted List: 943

The tree is not a max heap, so heapify. Swap 73 and 845, 73 and 645, and finally 73 and 243.

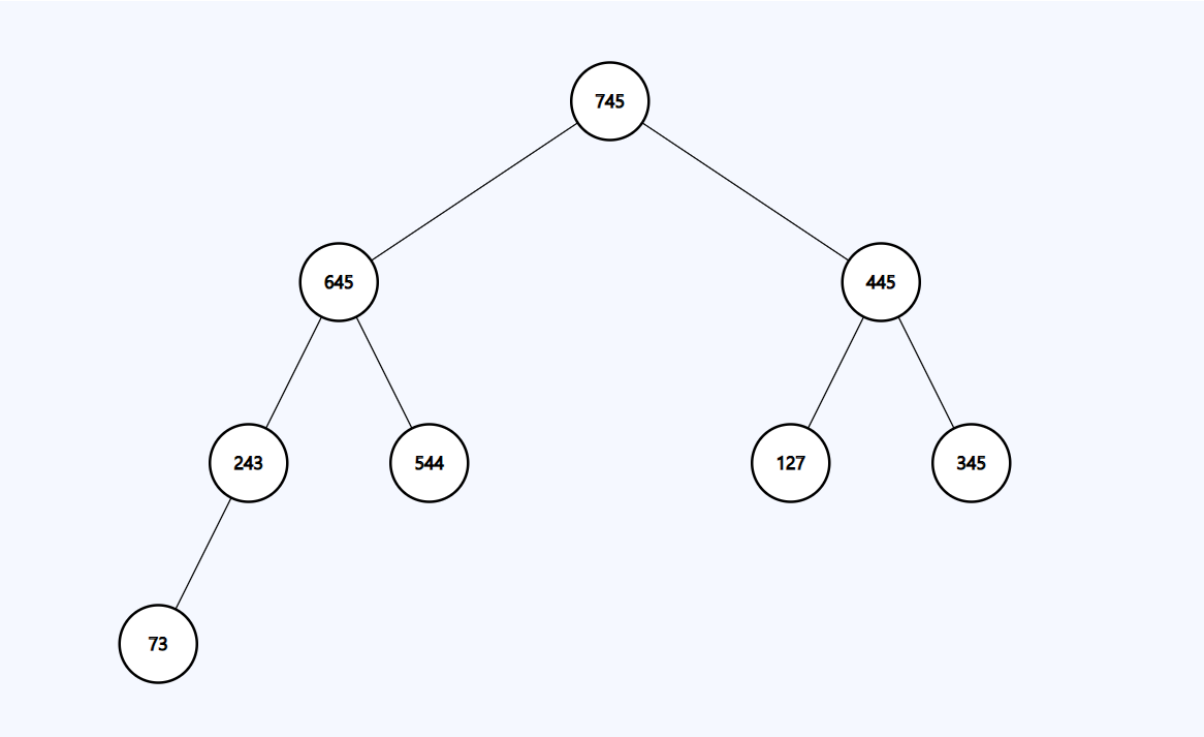


Now, swap 845 and 127, and remove 845.

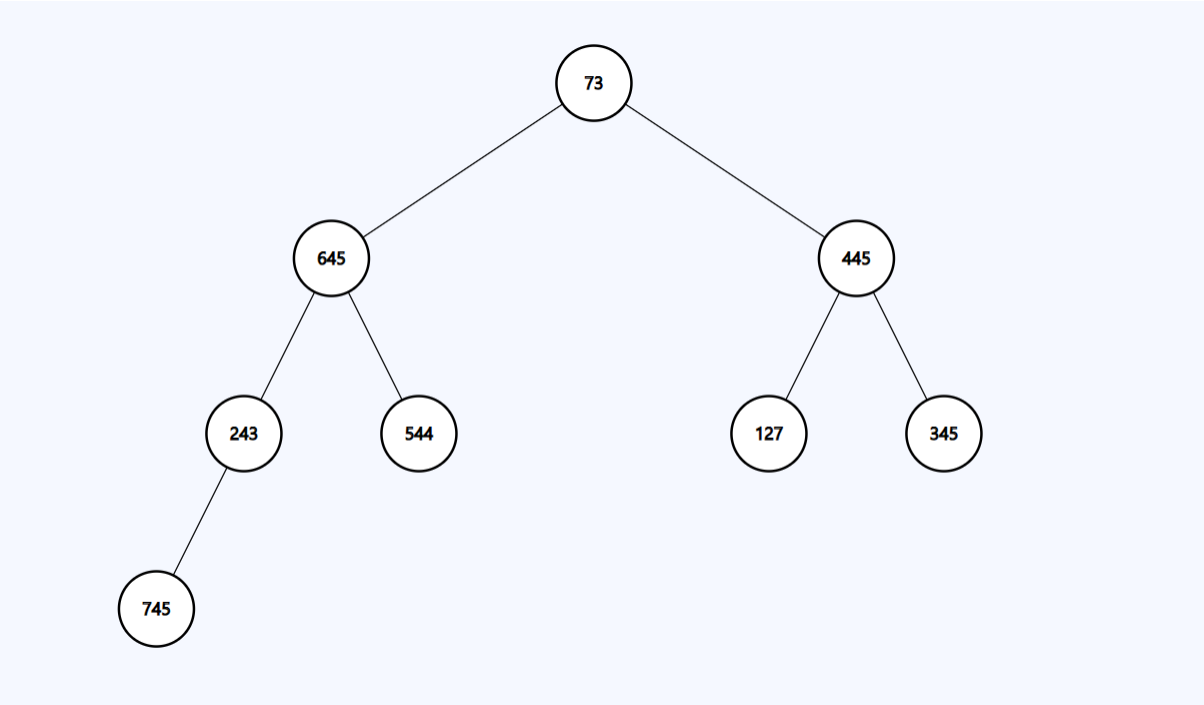


Sorted List: 845, 943

Heapify the tree again into a max heap. Swap 127 and 745, then 127 and 445.

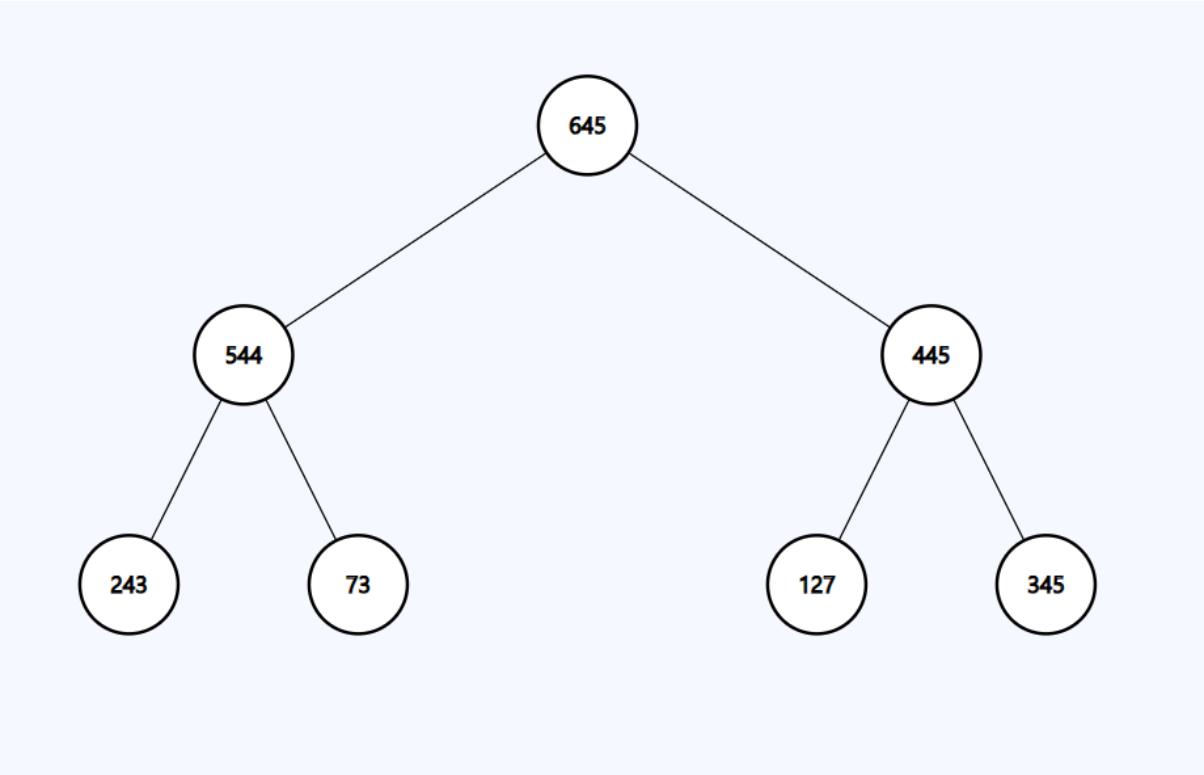


Swap 745 and 73, then remove 743.

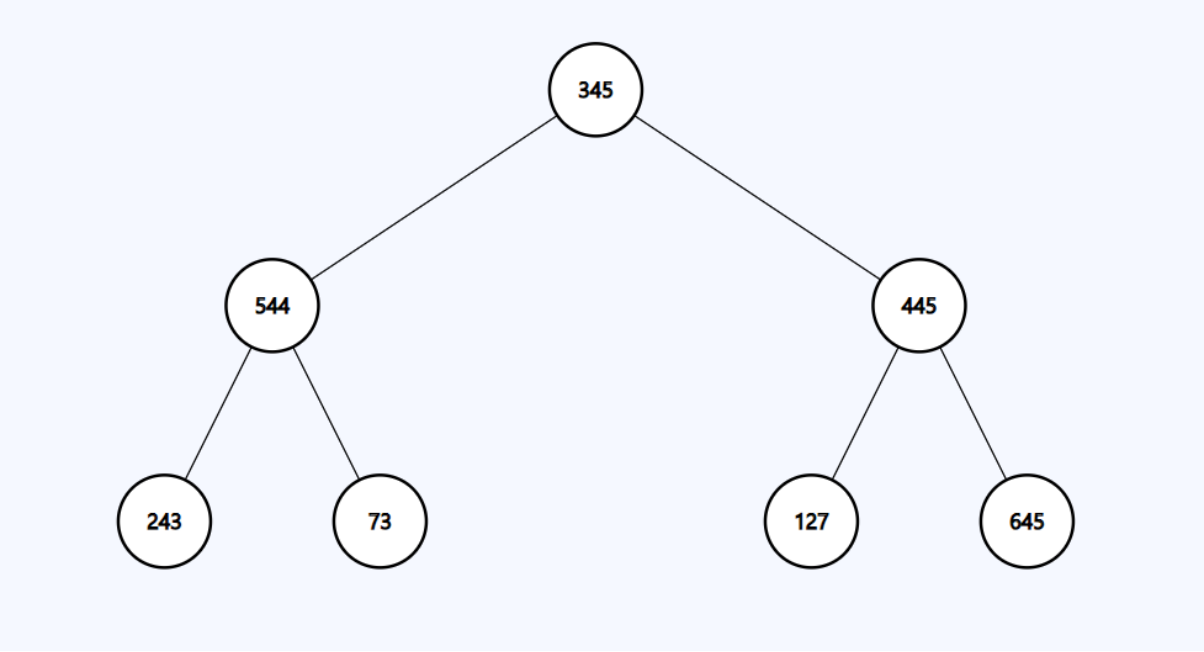


Sorted List: 745, 845, 943

To heapify, swap 73 and 645, then 73 and 544.

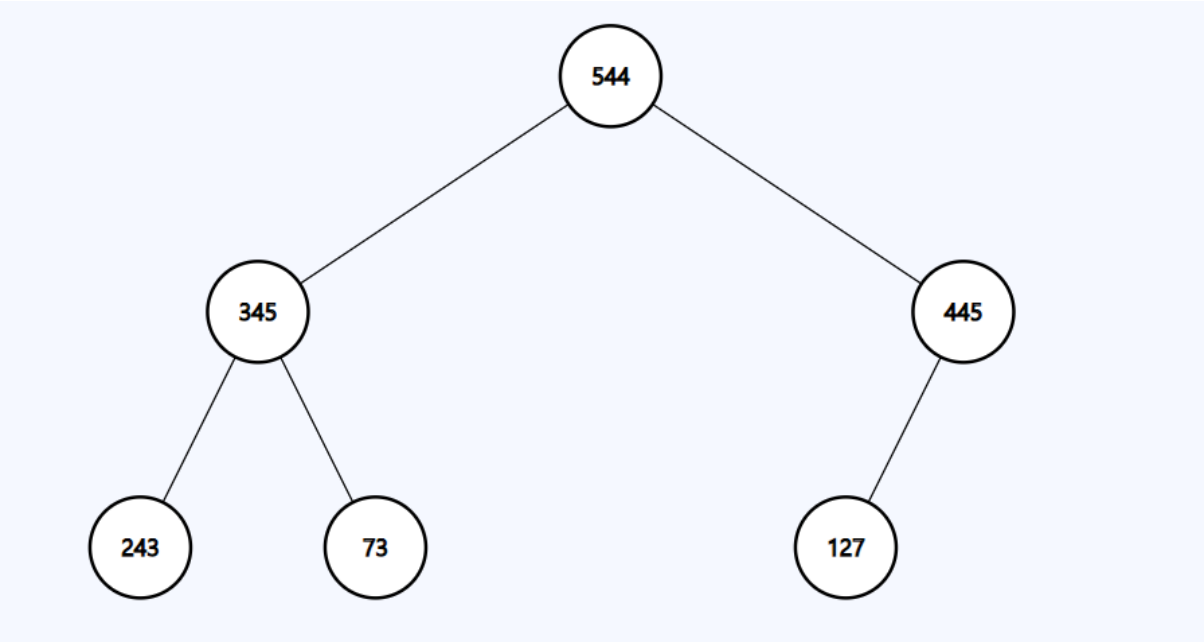


Swap 645 and 345, then remove 345.

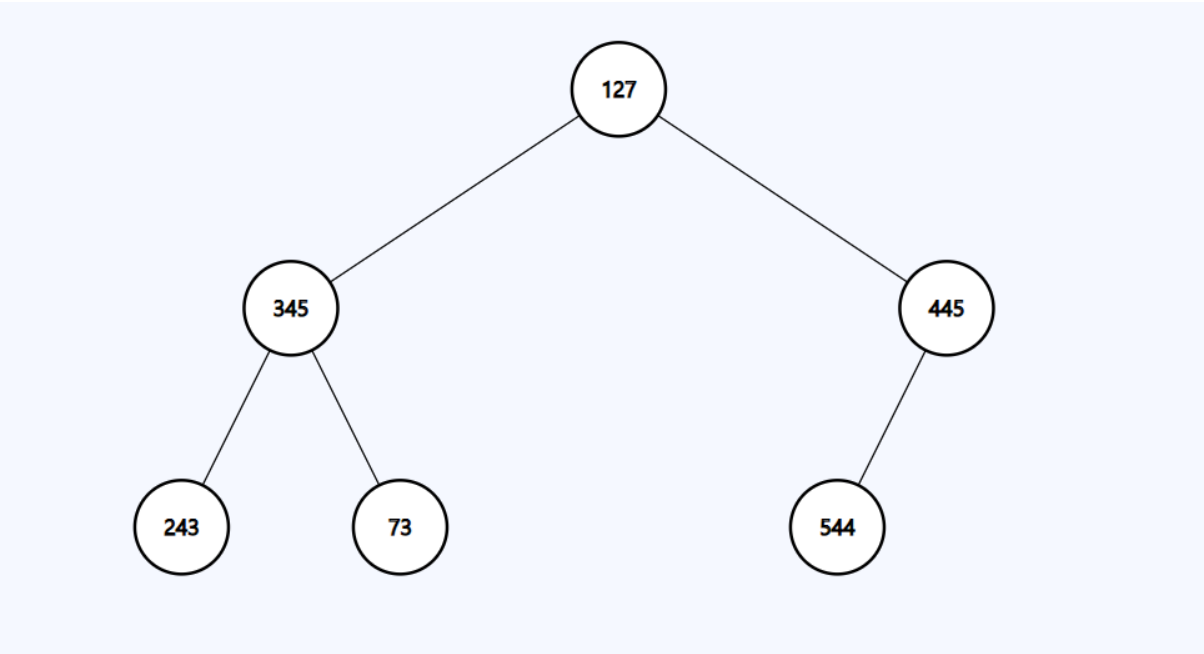


Sorted List: 645, 745, 845, 943

Swap 345 and 544 to heapify.

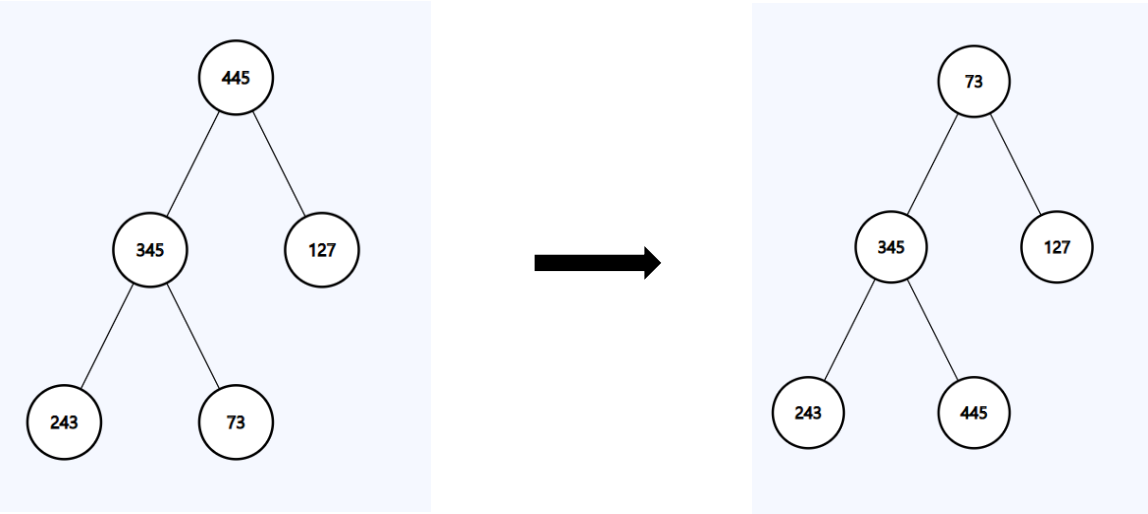


Swap 544 and 127, then remove 544.



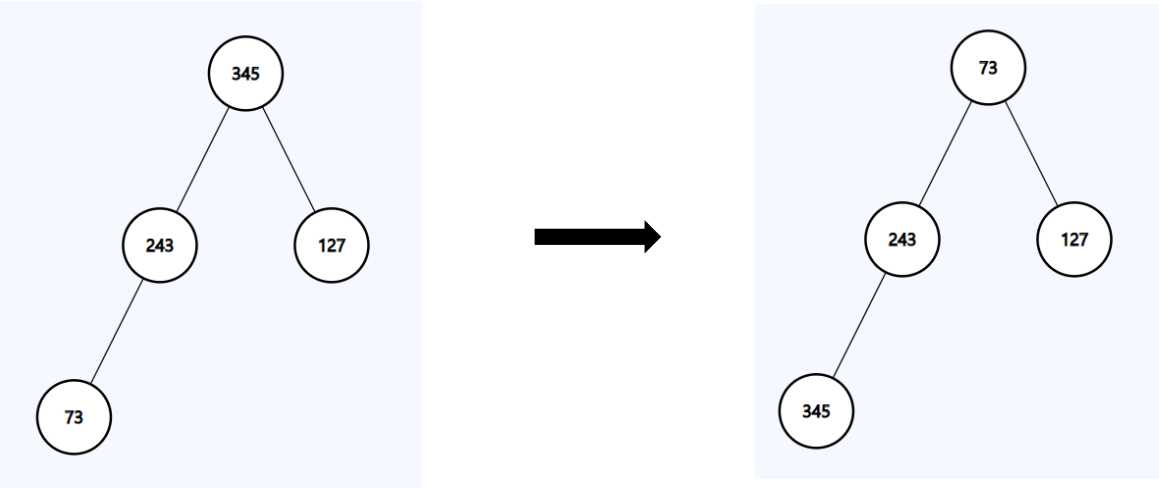
Sorted List: 544, 645, 745, 845, 943

To heapify, swap 127 and 445. Then, swap 445 and 73, and remove 445.



Sorted List: 445, 544, 645, 745, 845, 943

Swap 73 and 345, then 73 and 244 in order to heapify. Now, swap 345 and 73, remove 345.

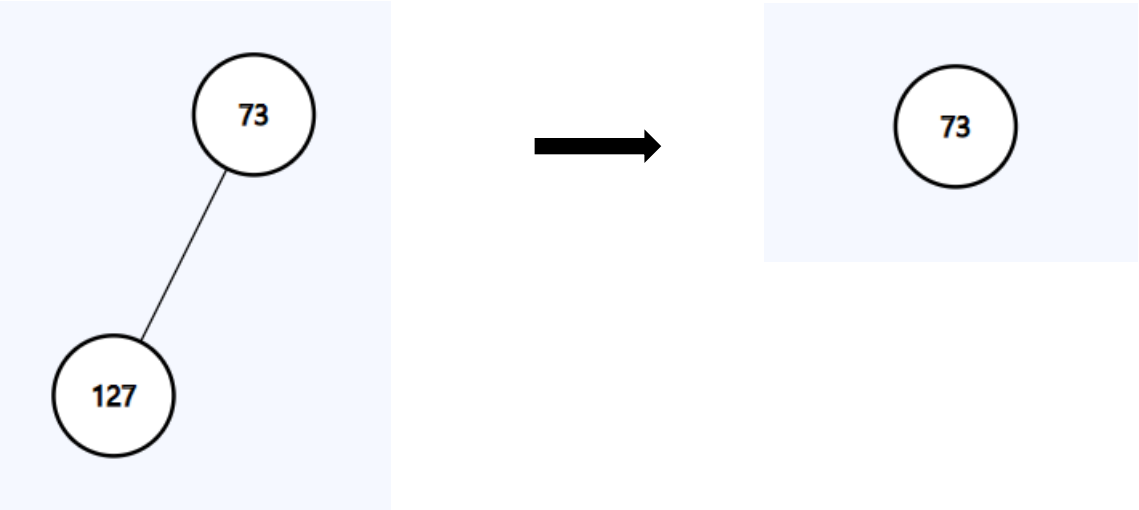


Sorted List: 345, 445, 544, 645, 745, 845, 943

To heapify, swap 73 and 243. Then, swap 243 and 127, and remove 243.

Sorted List: 243, 345, 445, 544, 645, 745, 845, 943

The tree is already a max heap. So now, swap 127 and 73, and remove 127. There is only one node remaining, which is 73. Remove that as well.



Final Sorted List: 73, 127, 243, 345, 445, 544, 645, 745, 845, 943

Heap Sort of 10 numbers has been successfully implemented.